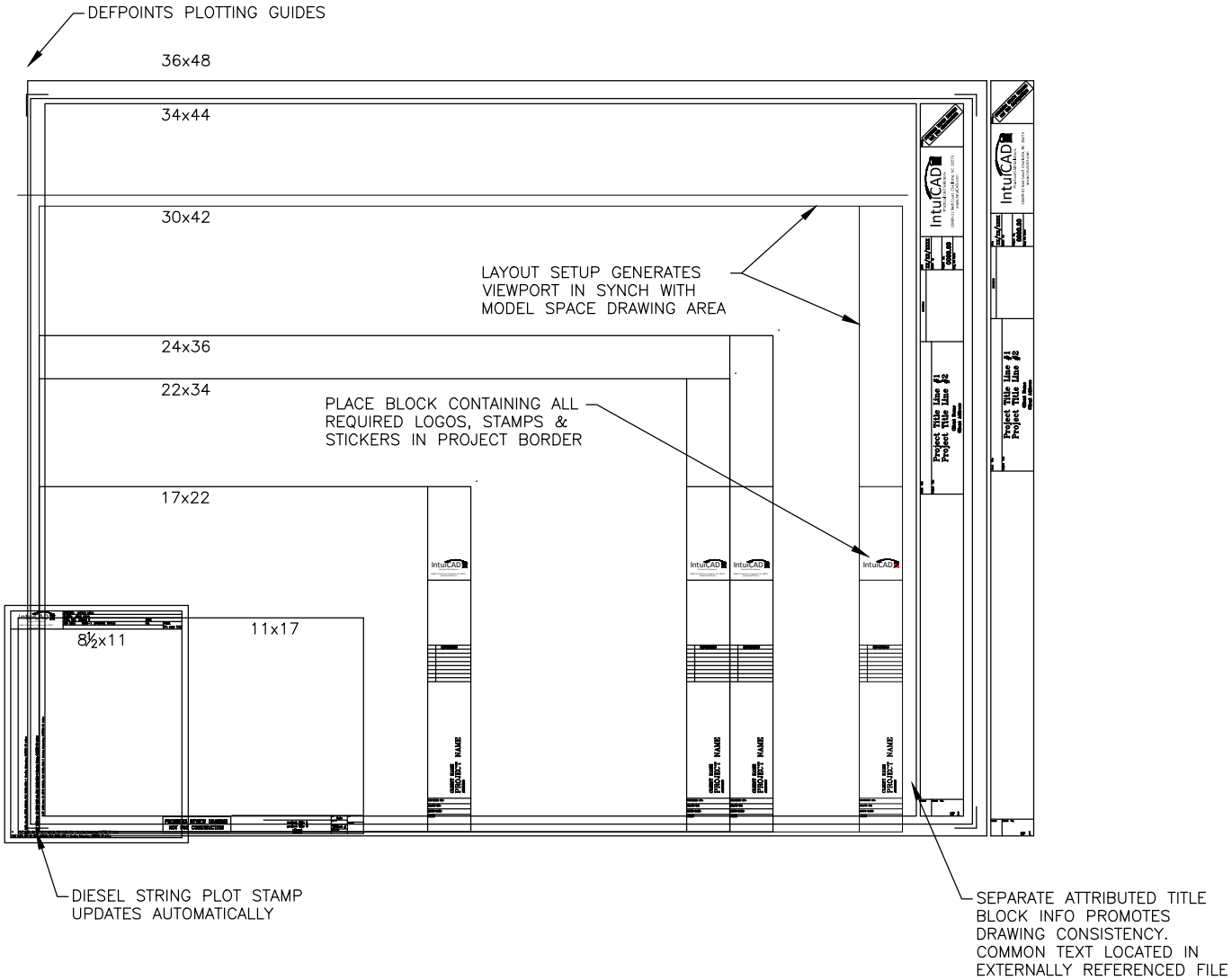
IntuiCAD

Structural Component Manager 2010 Product Manual

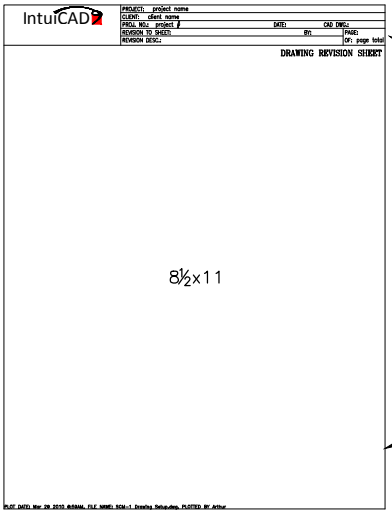


INTUICAD SCM USES THE CRITICAL POINT OF DRAWING CREATION TO IMPLEMENT NUMEROUS PREPARATIONS THAT WILL ALLOW THE MOST PRODUCTIVE USE AND MANIPULATION OF THAT DRAWING. MACHINATIONS INVOLVING A DOZEN DIFFERENT COMMANDS, TASKS AND SETTINGS ARE REDUCED TO A SINGLE CLICK.

THE LAYOUT AND MODEL SETUP ROUTINES CAN COMPLETELY AUTOMATE ALL DRAWING SETUPS IN THE OFFICE AND EXECUTE IN MERE SECONDS. WITH THE CLICK OF A BUTTON, THEY PROVIDE:

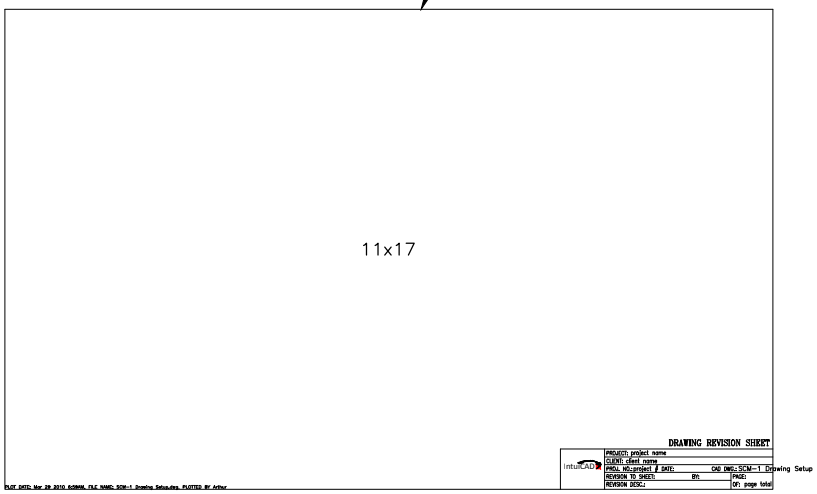
- CHOICE OF BORDER SIZE
- SET UP OF THE MODEL SPACE DRAWING SCALE
- ADDITION OF THE OFFICE STANDARD PRE–DEFINED OFFICE LAYERS
- ADDITION OF THE OFFICE STANDARD PRE–DEFINED TEXT STYLES
- ADDITION THE OFFICE STANDARD PRE–DEFINED DIMENSION TYPES
- INSERTION OR X–REF OF THE BORDER IN THE LAYOUT OR MODEL TAB
- CREATION OF A "DRAWING AREA" WINDOW IN THE MODEL TAB (LAYOUT ONLY)
- CREATION OF A VIEWPORT IN THE LAYOUT TAB ZOOMED TO SCALE (LAYOUT ONLY)
- IMPORT OF THE PAGE SETUP(S) INFORMATION FROM THE SELECTED BORDER.

WITH THE INTUICAD CENTRAL REPOSITORY STORING YOUR OFFICE STANDARDS, ALL DESIGNERS IN THE OFFICE WILL BE ACCESSING A SINGLE SOURCE FOR THE DRAWING SETUP ROUTINE; ALL DRAWINGS WILL BE CREATED THE SAME, BY EVERY DESIGNER, EVERY TIME. A TASK THAT TYPICALLY TAKES OVER 30 MINUTES, DOZENS OF STEPS AND IS PRONE TO INCONSISTENCIES; DRAWING TO DRAWING AND DESIGNER TO DESIGNER, NOW WILL TAKE 30 SECONDS OR LESS AND BE EXACTLY THE SAME. EVERY TIME.



REVISION SKETCH BORDERS ARE SET UP TO CREATE SEPERATE TABS FOR EACH REVISION SKETCH. THE TABS ARE LINKED TO THE ATTRIBUTES IN THE BORDERS.

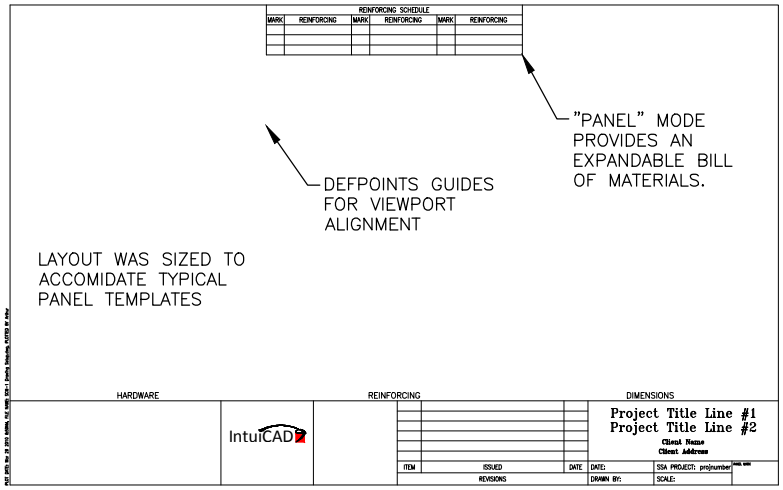
REVISION SKETCH BORDERS DO NOT ALTER DRAWING SETTINGS

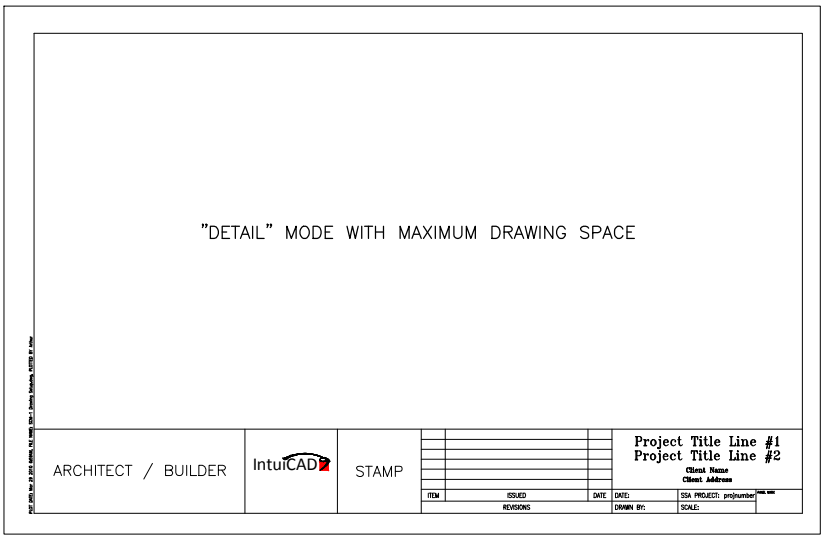


THE INTUICAD SCM REVISION SKETCH ROUTINE IS DESIGNED TO HANDLE THE SPECIAL REQUIREMENTS OF REVISION AND ADDENDUM MANAGEMENT. BY LOCATING AND PRE-EDITING PROJECT INFORMATION AT THE TIME OF PROJECT DIRECTORY CREATION AND EXTERNALLY REFERENCING THE BORDER, PROJECT CONSISTENCY CAN BE GUARANTEED. THIS METHOD, IN COMBINATION WITH DIESEL STRING TAB AND DRAWING NAME TRACKING CAN REDUCE TYPICAL EDITING BY 75-90% AND VIRTUALLY ELIMINATE ONE OF THE LEAST COST BENEFICIAL MARK-UPS IN ANY OFFICE.

THE REVISION SKETCH BORDER ROUTINE ALLOWS THE DESIGNER TO CHOOSE THE BORDER SIZE AND NAME THE LAYOUT TAB, THEN INSERTS OR X-REFS THE BORDER INTO THE NAMED TAB, CREATES A VIEWPORT AND ZOOMS THE MODEL EXTENTS. IT ALSO IMPORTS THE PAGE SETUP(S) INFORMATION FOR THE BORDER SELECTED.

OUR REVISION SKETCH ROUTINE CAN PROVIDE SIGNIFICANT SAVINGS AND SHORTEN RESPONSE TIME.





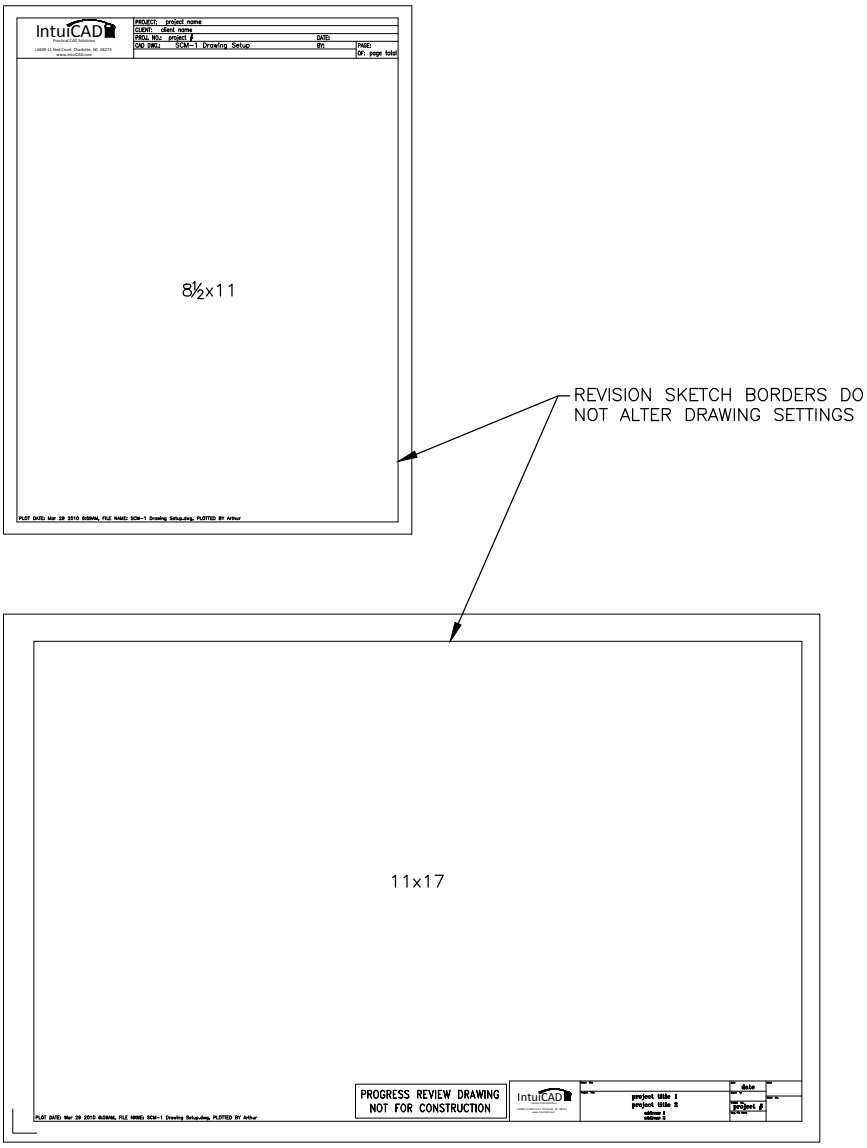
INTUICAD SCM USES THE CRITICAL POINT OF DRAWING CREATION TO IMPLEMENT NUMEROUS PREPARATIONS THAT WILL ALLOW THE MOST PRODUCTIVE USE AND MANIPULATION OF THAT DRAWING. MACHINATIONS INVOLVING A DOZEN DIFFERENT COMMANDS, TASKS AND SETTINGS ARE REDUCED TO A SINGLE CLICK AND PERFORMED FLAWLESSLY EVERY TIME.

THE TILT PANEL BORDER SETUP IS A TEMPLATE AND DEMONSTRATION OF SOME OF THE OTHER FEATURES THAT MAY BE ACCOMMODATED BY A SETUP ROUTINE. IN THIS CASE, A PANEL DRAWING BORDER IS INTEGRATED WITH A DETAIL BORDER FOR MAXIMUM VERSATILITY.

PANEL SELECTION WILL FUNCTION LIKE THE REVISION BORDER IN THAT NO ALTERATIONS TO THE CURRENT DRAWING WILL BE MADE OUTSIDE OF CREATING THE NEW TAB.

DETAIL SELECTION WILL FUNCTION LIKE THE LAYOUT SETUP AND CREATE THE ENTIRE DRAWING ENVIRONMENT WITH LAYERS, STYLES, ETC.

EVEN DRAWING SETUPS USED ONLY OCCASIONALLY BENEFIT GREATLY FROM AUTOMATION BY ELIMINATING THAT "GETTING RE–FAMILIAR WITH THE PROCESS" PERIOD AND ALLOWING GREATER USE OF PERSONNEL UNFAMILIAR WITH THE PROCESS.



THE QUICK SKETCH BORDER ROUTINE WAS DESIGNED TO PROVIDE PROFESSIONALLY FRAMED SKETCHES IN MINIMAL TIME. A PORTION OF DRAWING CAN BE COPIED AND PRESENTED TO CLIENTS OR OTHER CONSULTANTS IN SECONDS.

THE USE OF DIESEL STRING ATTRIBUTES CAN FACILITATE EDITING AND PROVE USEFUL FOR PART-TIME CAD USERS. THE QUICK SKETCH BORDER ROUTINE ALTERS NO DRAWING SETTINGS AND ONLY ALLOWS A CHOICE OF BORDER SIZE.

THE ROUTINE INSERTS OR X-REFS THE SELECTED BORDER IN THE CURRENT TAB AT THE CURRENT DIMSCALE SETTING AND IMPORTS THE PAGE SETUP(S) INFORMATION FROM THE BORDER SELECTED.

MORE IMPORTANT THAN WHAT THE LAYERS ARE IS HOW THEY ARE PERCIEVED BY THE CREATOR OF THIS PACKAGE. LAYERS ARE FIRST AND FORMOST THE DRAFTER'S TOOL. A LAYERING SYSTEM SHOULD IDEALY BE DESIGNED TO ACCOMPLISH MORE BUT MUST FIRST FACILLITATE THE DRAFTING EFFORT.

THE LAYERS IN THIS SYSTEM SORT DRAWING ELEMENTS BY TYPE OF OBJECT. THIS PACKAGE ALSO MAKES USE OF THE SPECIAL CHARACTERISTICS OF LAYER 0 AND ASSIGNING COLOR AND/OR LINETYPE "BYBLOCK". A SIMPLE ADJUSTMENT TO THE SPECIFIC BLOCK WILL TUNE THIS TO THE CLIENT'S PREFERENCES.

ALL OF THE ITEMS WILL SELF ASSIGN LAYERS IF THE MENU OR TOOLBAR COMMAND IS USED. THE SEQUENCE OF DECISION IS AS FOLLOWS:

1. CHECK FOR THE LAYER THE OBJECT WILL BE ASSIGNED TO
2. IF THE LAYER EXISTS, SWITCH TO IT TEMPORARILY FOR THE COMING OPERATION THEN BACK TO THE PREVIOUS.
3. IF THE LAYER DOES NOT EXIST, PERFORM THE COMING OPERATION ON THE CURRENT LAYER.

THIS PROCESS ALLOWS THE COMPANY TOOLS TO BE USED IN OTHER CAD STANDARDS WITHOUT LAYER CREATION. THE FULL LIST OF LAYERS USED IS:

0	BLOCK CREATION LAYER, BORDER INSERTION LAYER, NOT USED FOR GENERAL DRAFTING
DEFPOINTS	USED BY AUTOCAD FOR DIMENSION DEFINITION POINTS, DOES NOT PLOT--USED FOR GUIDELINES & VIEWPORTS
S-ABLT	BOLTS, ANCHORS AND FASTENERS OF ALL TYPES
S-ARCH	ARCHITECTURAL (BACKGROUND) LINEWORK
S-BLDG	BUILDING OUTLINE
S-BRDG	JOIST BRIDGING
S-BRIK	BRICK
S-CMUW	CMU WALLS
S-CONC	CONCRETE EDGES ON PLAN
S-CONC-BEAM	CONCRETE BEAMS ON PLAN
S-CONC-COLS	CONCRETE COLUMNS ON PLAN
S-CONC-JOIS	CONCRETE JOISTS ON PLAN
S-DECK	STEEL DECK
S-DIMS	DIMENSIONS
S-EXST	LAYER FOR MINIMAL EXISTING STRUCTURE
S-EXST-XXXX	LAYERS FOR DRAWING EXISTING STRUCTURE WITH SHADED PENS AND VARIED LINEWIDTH
S-GRID	CENTERLINES
S-HTCH	HATCHING AND SHADING
S-MESH	WELDED WIRE FABRIC
S-MTCH	MATCH LINES
S-MECH	MECHANICAL EQUIPMENT OUTLINES
S-OPNG	OPENINGS
S-PIER	PIERS AND FOOTINGS
S-RBAR	REINFORCING BARS, WATERSTOPS
S-REVS	REVISION CLOUDS
S-REV_	REVISION DELTA GENERATED LAYERS
S-SECT-CONC	CONCRETE EDGES IN SECTION
S-SECT-STEL	STEEL ON SECTIONS
S-SLAB-JNTS	SLAB CONTROL JOINTS ON PLAN
S-STEL-BEAM	STEEL BEAMS AND GIRDERS ON PLAN
S-STEL-COLS	STEEL COLUMNS ON PLAN
S-STEL-JOIS	OPEN WEB STEEL JOISTS ON PLAN (CHANGE LAYER LINETYPE FOR SECTIONS)
S-TAGS	CROSS REFERENCE, TITLES, SYMBOLS AND CODED NOTES
S-TEXT	GENERAL DRAWING TEXT, TEXT-LIKE BLOCKS
S-WALL	WALLS OF VARIOUS DESCRIPTION
S-WELD	WELDING SYMBOLS
S-WOOD	WOOD, LIGHT GAGE STEEL, ACCESSORIES

"RESTORING" LAYERS TO OFFICE STANDARD OVERWRITES ANY ADJUSTMENTS TO LAYERS THAT MAY HAVE BEEN MADE PREVIOUSLY.

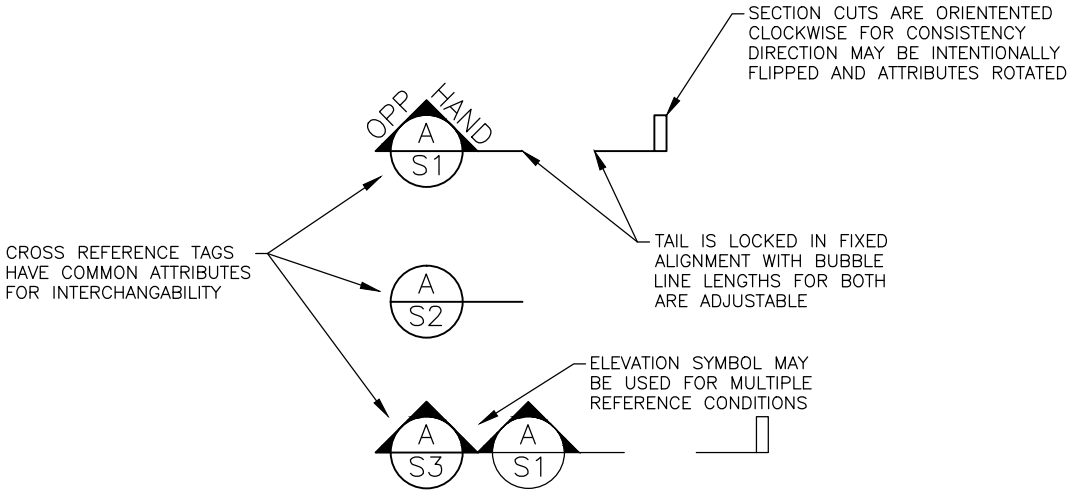
"ADDING" LAYERS NOT EXISTING DO NOT ALTER ANY LAYERS ALREADY PRESENT IN THE DRAWING.

"ADDING EXISTING" LAYERS WILL ADD OFFICE OBJECT LAYERS PRECEDED BY "S-EXST" INSTEAD OF "S-". THIS PROCESS DOES NOT ALTER ANY OBJECT'S LAYER STATE. THE NUMBERS FOR THE LAYERS WERE SELECTED TO MIRROR THE BLOCK OF NUMBERS ACTIVELY USED BUT WITH A 50% SCREENING APPLIED. THIS ALLOWS SHADED OBJECTS TO HAVE LINE WIDTH AND MAKES THE CONVERSION TO FOLLOW WORK EFFICIENTLY WITH MINIMAL ADDITIONAL EDITING TO THE DRAWING (EXCEPT ERASING).

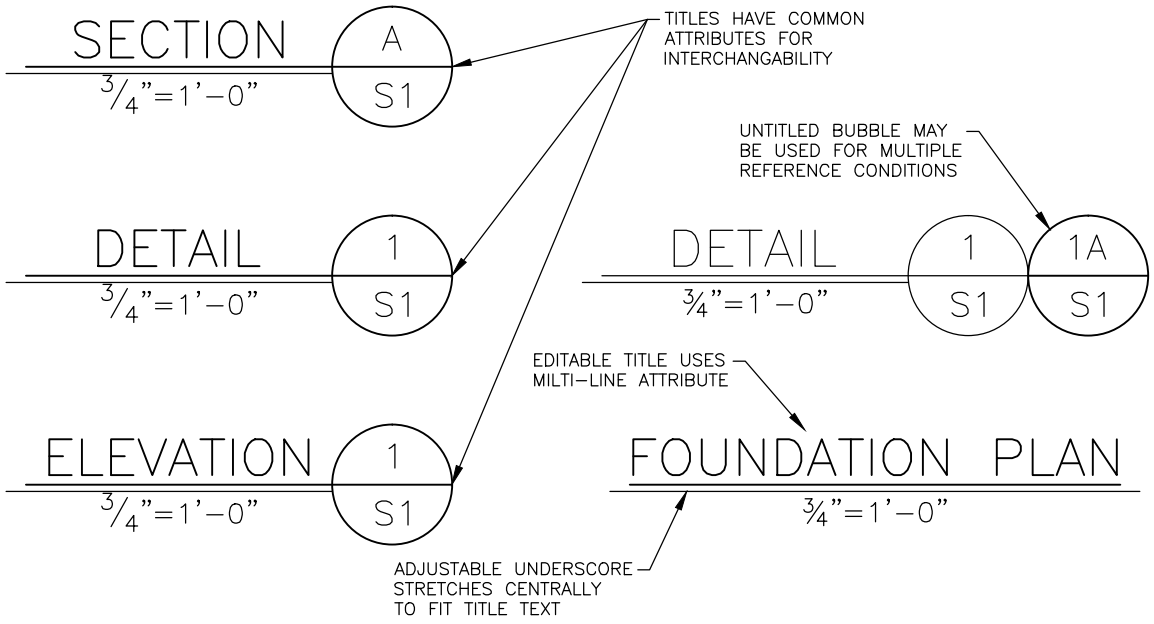
"CONVERTING LINEWORK" TO EXISTING LAYERS IS A PROCESS BY WHICH A DRAWING DONE WITH STANDARD OFFICE LAYERING IS CONVERTED TO A BACKGROUND DRAWING FOR ADDITIONS OR RENOVATION. EACH OFFICE OBJECT LAYER WILL HAVE IT'S CORRESPONDING EXISTING LAYER CREATED AND ALL OF IT'S OBJECTS TRANSFERRED TO IT. THIS PROCEDURE IGNORES "NON-OBJECT" LAYERS SUCH AS GRID, TEXT AND TAG LAYERS AND ASSUMES ANY HATCHING WILL BE ROMOVED AND IGNORES IT AS WELL. ANY OBJECTS WITH COLORS ASSIGNED MUST BE HANDLED INDIVIDUALLY. THIS PROCEDURE ALSO WORKS WELL WHEN EXTENSIVE DRAWINGS OF EXISTING STRUCTURES MUST BE CREATED. THE BUILDING CAN BE DRAWN AS USUAL AND BE CONVERTED PRIOR TO ADDING NEW STRUCTURE. THE ROUTINE WILL NOT WORK IF EXISTING LAYERS ALREADY EXIST.

A FULL SET OF DIMENSION STYLES CAN BE CREATED TO USER SPECIFICATIONS FOR ANY DIMENSIONING METHOD. INTUICAD'S ENHANCED DIMENSIONING BENEFITS CONSISTS SOLELY OF HAVING A BUTTON CAPABLE OF RESTORING THEM TO A PREDETERMINED STATE WHEN NECESSARY. THE STYLE NAME "STANDARD" HAS SPECIAL FUNCTIONAL CAPABILITIES TO AUTOCAD.

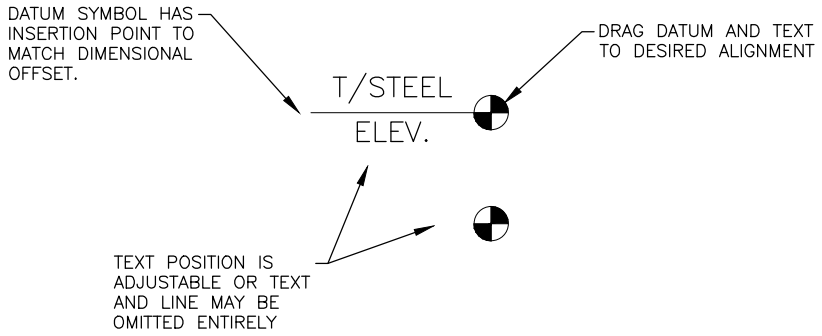
THERE ARE TWO TEXT STYLES IN THE SYSTEM, "STANDARD" AND "TITLE". THE "STANDARD" NAME IN THIS CASE ALLOWS THE BLOCK ATTRIBUTES TO ADAPT TO ALTERNATE FONTS. THE "TITLE" ALLOWS FOR A SECOND FONT (IF DESIRED) IN TITLES AND SCHEDULES.



NOT ONLY HAVE OUR DYNAMICS ADDRESSED THE TEXT ORIENTATION ISSUES OF SECTION MARKS AS THEY ARE ROTATED, WE HAVE COMBINED THEM WITH OTHER PLAN CROSS REFERENCE TAGS TO ALLOW THE USER IMMEDIATE ACCESS TO THEIR CHOICE. ATTRIBUTES ARE SHARED TO MINIMIZE EDITING AFTER THEY ARE COPIED.

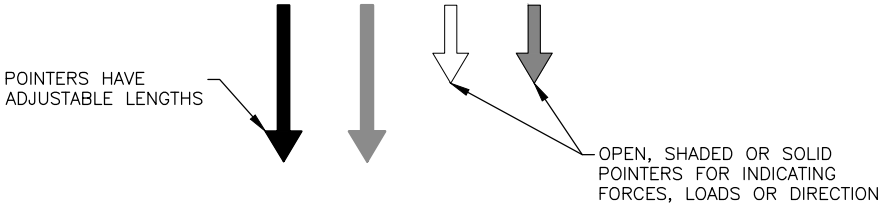


INTUICAD SCM'S COMBINED TITLES ALLOW AN OFFICE TO CONTROL OR ALLOW EDITING FOR STANDARD TITLES. ATTRIBUTES ARE SHARED AND SAVE EDITING DURING THE SHEET BUILDING PROCESS.



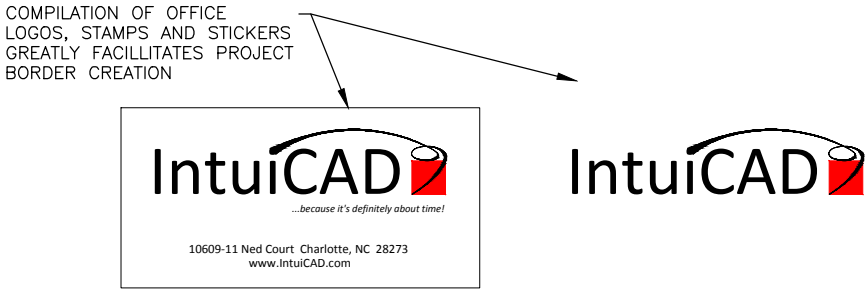
THE DATUM SYMBOL IS DESIGNED FOR FAST ALIGNMENT ON SECTIONS AND ELEVATIONS.

THE DATUM SYMBOL IS INSERTED ON THE S-DIMS LAYER WITH LINEWORK BYLAYER AND TEXT COLOR ASSIGNED.



THE POINTER SYMBOL PROVIDES OPTIONS ON LOAD DIAGRAMS AND TRUSS ELEVATIONS.

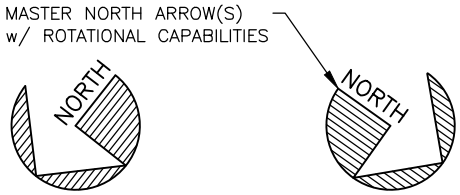
THE POINTER SYMBOL IS INSERTED ON THE TAGS LAYER WITH COLOR BYBLOCK. THE SHADING COLOR IS ASSIGNED WITHIN THE BLOCK. THE SOLID COLOR IS BYBLOCK, THEREFORE ASSIGNING SHADED COLOR AFTER INSERTING WILL MAKE LINES "INVISIBLE".



THE LOGO PROVIDED SERVES AS A TEMPLATE FOR COMPANY LOGO CREATION.

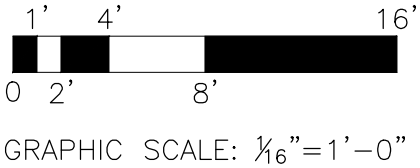
THE LOGO SHOULD BE CREATED WITH ALL COLORS ASSIGNED AND A CENTRALLY LOCATED INSERTION THAT FACILITATES BORDER PLACEMENT.

THE LOGO IS INSERTED ON LAYER 0 AND IS SCALED TO THE DIMSCALE.

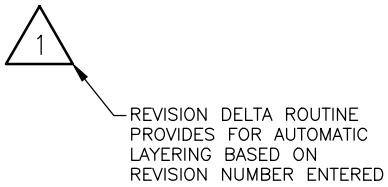


THE NORTH ARROW VARIES WILDLY FROM OFFICE TO OFFICE. THIS ONE WAS SELECTED BECAUSE OF JUSTIFICATION REQUIREMENTS. THE DUAL VISIBILITY MODES ALLOW THE TEXT TO MAINTAIN READING ORIENTATION. SYMMETRICAL NORTH ARROWS AND THOSE WITHOUT TEXT ARE FAR EASIER.

THE NORTH ARROW HEAVY LINEWORK AND TEXT ARE BYLAYER. THE LIGHT LINES ARE ASSIGNED.

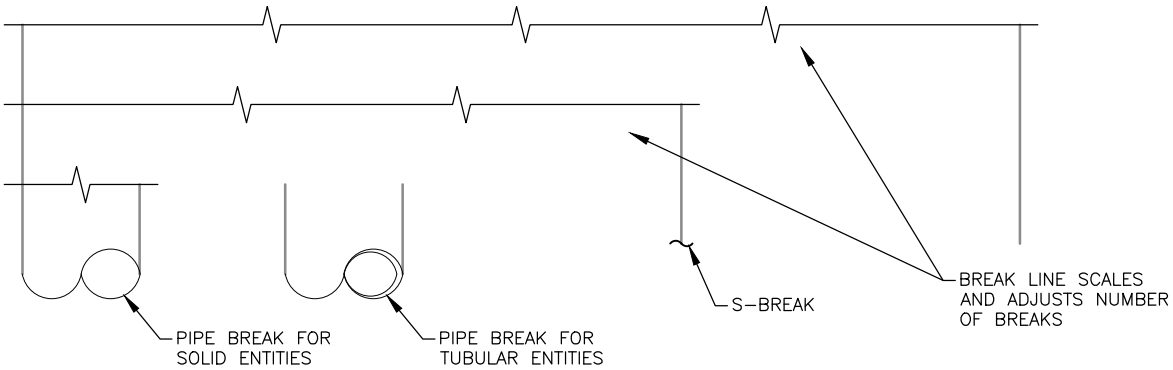


GRAPHIC SCALES ARE INSERTED ON THE "S-TEXT" LAYER WITH THE LINEWORK COLOR ASSIGNED. THEY ARE INSERTED WITH A SACLE OF 1.



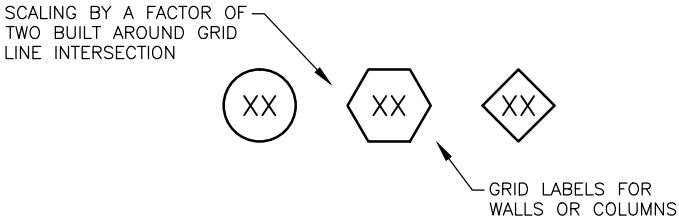
THE REVISION DELTA ROUTINE HAS A BUILT IN AUTOMATION THAT WILL CREATE A LAYER NAMED "S-REV_" WHERE "_" IS THE VALUE ENTERED INTO THE REVISION DELTA. THIS IS ONLY TRIGGERED WHEN THE S-REV0 LAYER IS PRESENT ALREADY. TO DISABLE THIS FEATURE (WHILE RETAINING LAYER S-REV0) SIMPLY ENTER A RETURN AT THE REVISION NUMBER PROMPT AND THE DELTA WILL BE INSERTED ON THE CURRENT LAYER.

THE REVISION DELTA LINEWORK IS BYLAYER. THE TEXT COLOR IS ASSIGNED.

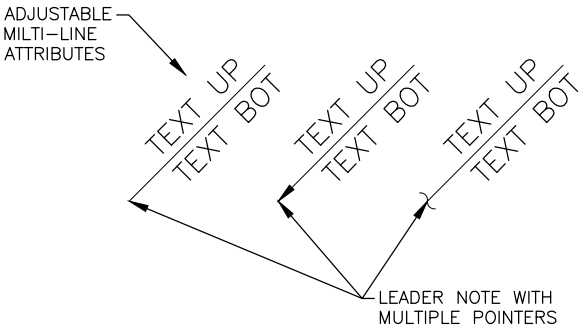


INTUICAD SCM BREAK ROUTINES PROVIDE BOTH SPEED AND UNIFORMITY AND REDUCE "TRIMMING UP" AN ENTIRE SHEET OF SECTIONS TO A FLURRY OF MOUSE CLICKS. BREAK LINES AND PIPE BREAKS ARE PARAMETRIC ROUTINES THAT DRAW LINEWORK ON THE CURRENT LAYER, USING CURRENT LINETYPE AND COLOR SETTINGS.

THE S-BREAK IS INSERTED ON THE CURRENT LAYER WITH THE COLOR BYBLOCK.

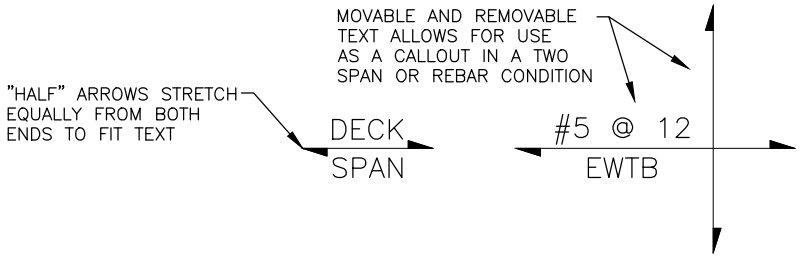


THE GRID LABEL LINEWORK COLOR IS ASSIGNED DURING INSERTION AND THE ATTRIBUTE COLOR IS BYLAYER. THE GRID LABEL LINEWORK LINETYPE IS BYBLOCK.

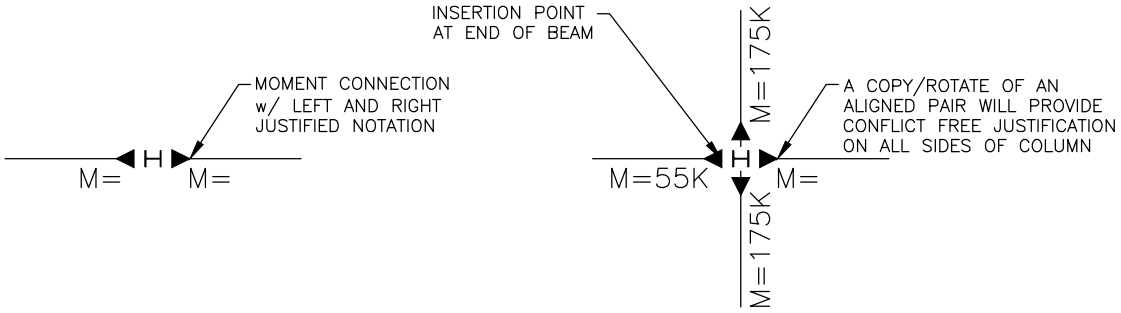


THE LEADER NOTE IS AN ALL PURPOSE LABEL FOR COLUMNS, FOOTINGS OR PIERS, DEPENDING ON OFFICE PRACTICES. AT THE LEAST, IT IS A PLATFORM FOR OTHER BLOCKS. IF NOT FOR THE EXISTING VISIBILITY PARAMETER EXCHANGING ARROWHEADS, TAILS AND ADDITIONAL NOTATION COULD EASILY BE ADDED TO CREATE A "SUPER TAG".

THE LEADER NOTE IS INSERTED ON THE "S-TEXT" LAYER WITH THE LINEWORK COLOR ASSIGNED UPON INSERTION.

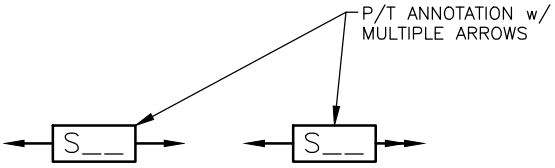


THE SPAN INDICATOR INSERTS ON THE "S-TEXT" LAYER WITH THE LINEWORK ASSIGNED.



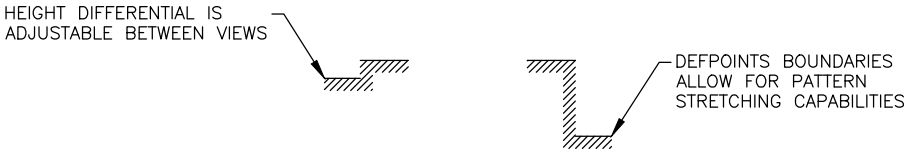
THE MOMENT CONNECTION SYMBOL WAS DESIGNED SPECIFICALLY TO REMOVE THE "ADJUST TEXT" FOLLOW UP TO MANY EDITS. THE JUSTIFICATIONS ALLOW THE TEXT TO BE EDITED WITHOUT THE SECONDARY ACTION.

THE MOMENT CONNECTION SYMBOL IS INSERTED ON THE "S-STEL-BEAM" LAYER FOR CONVENIENCE OF VIEWING (AND EDITING) FORCES WHILE THE TEXT LAYER MAY BE OFF OR FROZEN. IT CONVERSELY ALSO ALLOWS THE FORCES TO BE TURNED OFF WHEN THE FRAMING LINEWORK IS NOT AVAILABLE.



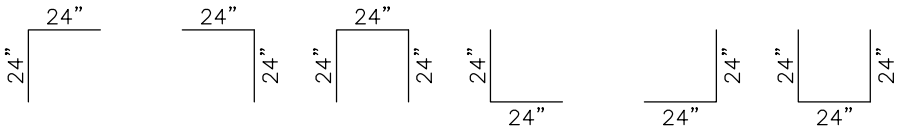
THE POST TENSION SYMBOL COMBINES THE FLEXIBILITY OF AN ASSOCIATIVE DIMENSION WITH BOLD LINEWORK WHILE ALLOWING QUICK EDITING OF ARROW ENDS.

THE POST TENSION SYMBOL LINEWORK IS ASSIGNED WITHIN THE BLOCK AND THE TEXT IS ASSIGNED DURING INSERTION.



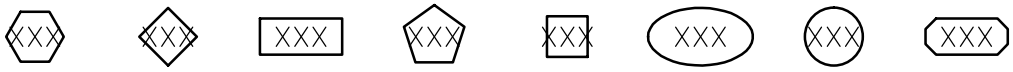
THE DEPRESSION INDICATOR HAS A VISIBILITY "REVERSE" INSTEAD OF A "FLIP" ACTION IN ORDER TO MAINTAIN CONSISTENT HATCH DIRECTION.

THE DEPRESSION SYMBOL HATCH IS ASSIGNED WITHIN THE BLOCK. THE VISIBLE LINEWORK COLOR IS ASSIGNED UPON INSERTION.



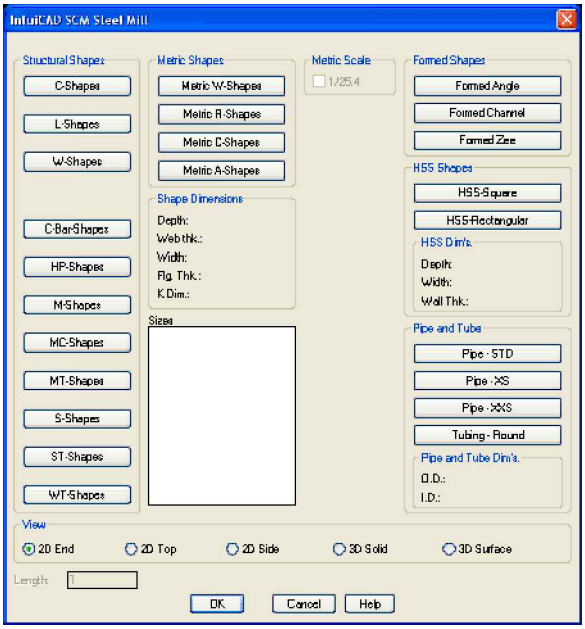
THE BAR BEND/SLOPE NOTE IS DESIGNED TO BE EXPLODED USING AUTOCAD'S BURST COMMAND (OR THE BONUS "LX" ROUTINE) UPON SELECTION OF THE DESIRED NOTE.

THE BAR BEND/SLOPE NOTE IS NOT EDITABLE AS A BLOCK.



THE CODED NOTES BLOCK HAS SYNCHRONIZED ATTRIBUTES THAT MAINTAIN THEIR VALUE AFTER THE SHAPE RE-SELECTED.

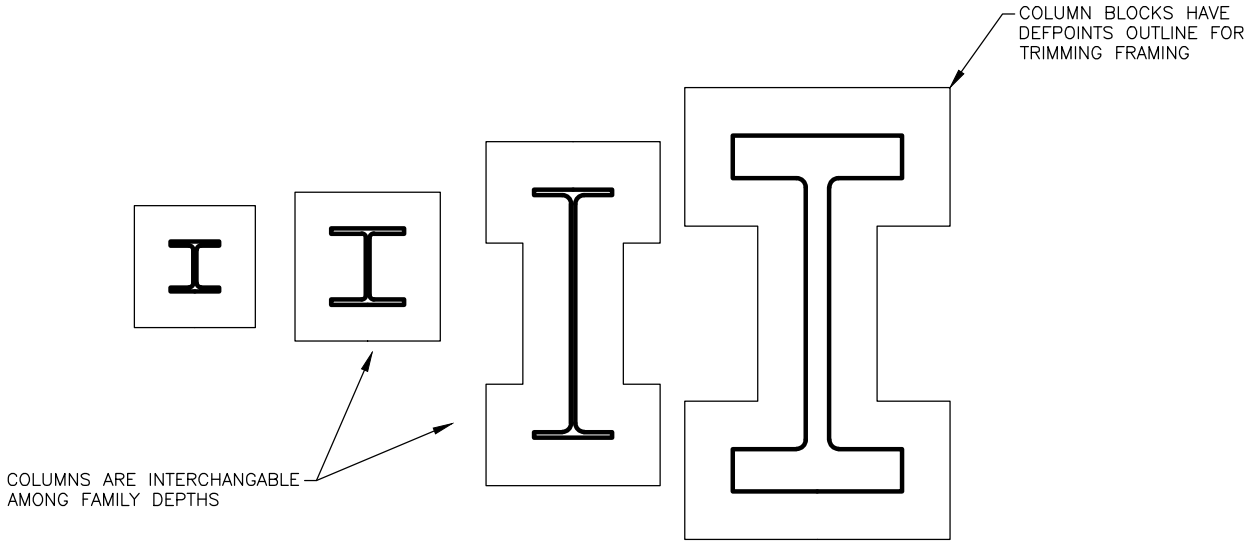
PARAMETRIC STEEL MILL GENERATES NEARLY ALL SHAPES ASTM AND METRIC



PARAMETRIC SHAPE ROUTINES HAVE LONG BEEN A FAVORITE TOOL FOR CAD CUSTOMIZATION BECAUSE OF THE FLEXIBILITY OF THE FINISHED LINEWORK. INTUICAD SCM STEEL MILL IS A PARAMETRIC SHAPE DRAWING PROGRAM THAT READS DATA FROM SUPPLIED FILES.

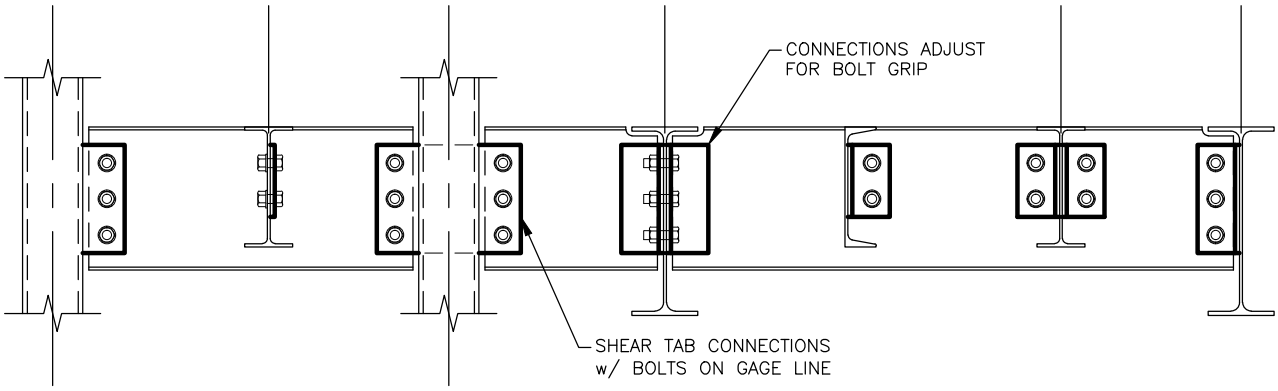
THE CONTENT IN THE DATA FILES MAY BE EXPANDED BY ADDING TO THEM FOLLOWING THE SAME FORMAT.

THE OPTIONS ON THE DIALOG ARE STRAIGHTFORWARD AND EASY TO FOLLOW.



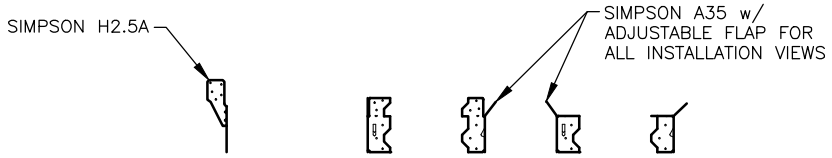
BLOCKED COLUMN SHAPES ARE FAR MORE EFFICIENT ON PLAN WORK OR WHEN EXPORTING TO OTHER DESIGN PROGRAMS. INTUICAD SCM BLOCKED COLUMNS ARE GROUPED BY FAMILIES WITH DEFPOINTS LINES FOR TRIMMING FRAMING MEMBERS. WITH TECHNIQUE, AN OFFICE COLUMN BLOCK CAN QUICKLY BECOME A PROJECT COLUMN BLOCK WITH HIDDEN OR SPLICED VIEWS.

THE BLOCKED COLUMNS ARE INSERTED ON THE "S-STEL-COLS" LAYER.

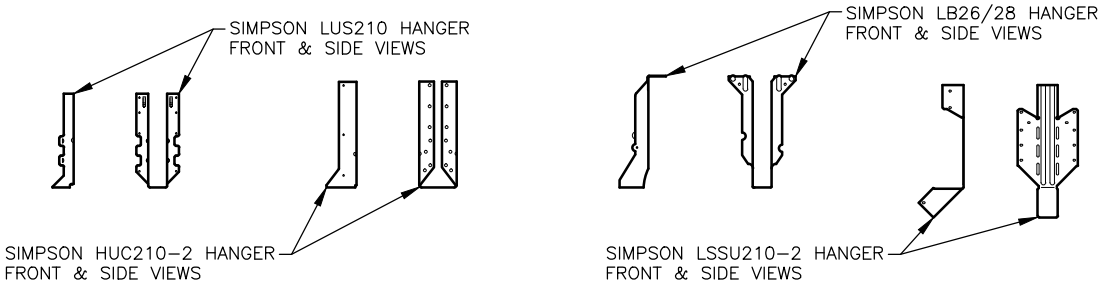


THE PLATE/ANGLE CONNECTOR PROVIDES AN ASSORTMENT OF FRAMING CONNECTIONS WITH SYNCHRONIZED DYNAMICS. THESE CONNECTIONS ARE HIGHLY RESPONSIVE TO ADJUSTMENTS IN FRAMING MEMBER SIZES, ALLOWING SECTIONS TO BE INTERCHANGED MORE FREELY WHILE MAINTAINING INDUSTRY STANDARD GAGES AND SPACINGS.

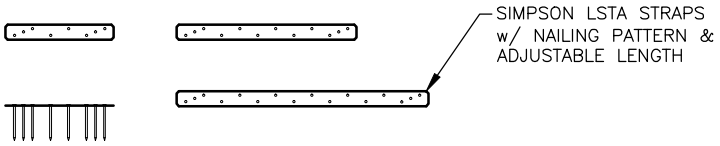
THE PLATE/ANGLE CONNECTOR LINEWORK IS BYLAYER WITH THE BOLTHEADS ASSIGNED. THE BLOCK IS ASSIGNED A HIDDEN LINETYPE.



GAGE CLIP ANGLES



GAGE HANGERS

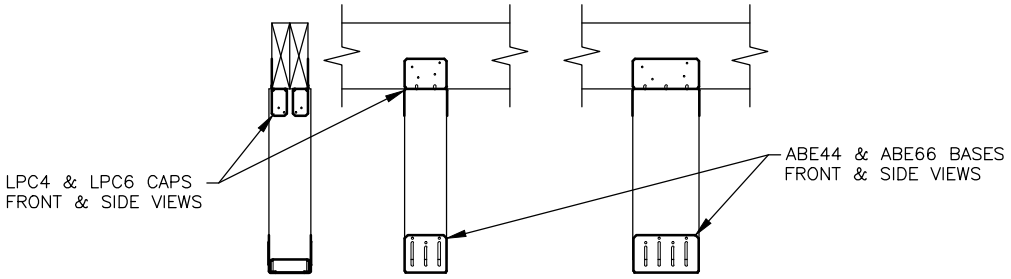


GAGE STRAPS

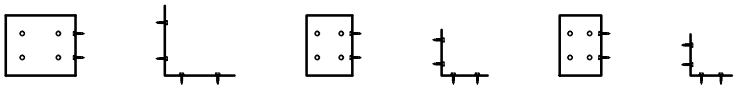
INTUICAD SCM GAGE CONNECTORS NOT ONLY PROVIDE QUICK ACCESS TO A LIBRARY OF TYPES BUT ALSO A PLATFORM TO ADD OTHERS THE OFFICE MAY UTILIZE FREQUENTLY.

GAGE CONNECTOR OUTLINE LINEWORK COLOR IS BYBLOCK WITH THE REMAINDER OF THE LINEWORK BYLAYER. THE FASTENERS ARE ASSIGNED TO THE "S-ABLT" LAYER WITHIN THE BLOCK.

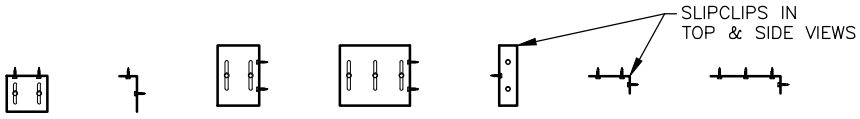
GAGE CLIP ANGLE BLOCKS ARE INSERTED ON THE "S-WOOD" LAYER.



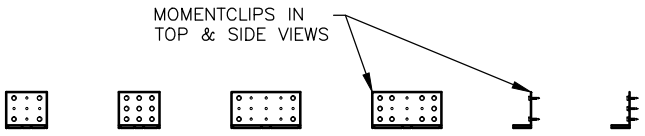
GAGE BASES & CAPS



STIFFCLIPS



SLIPCLIPS



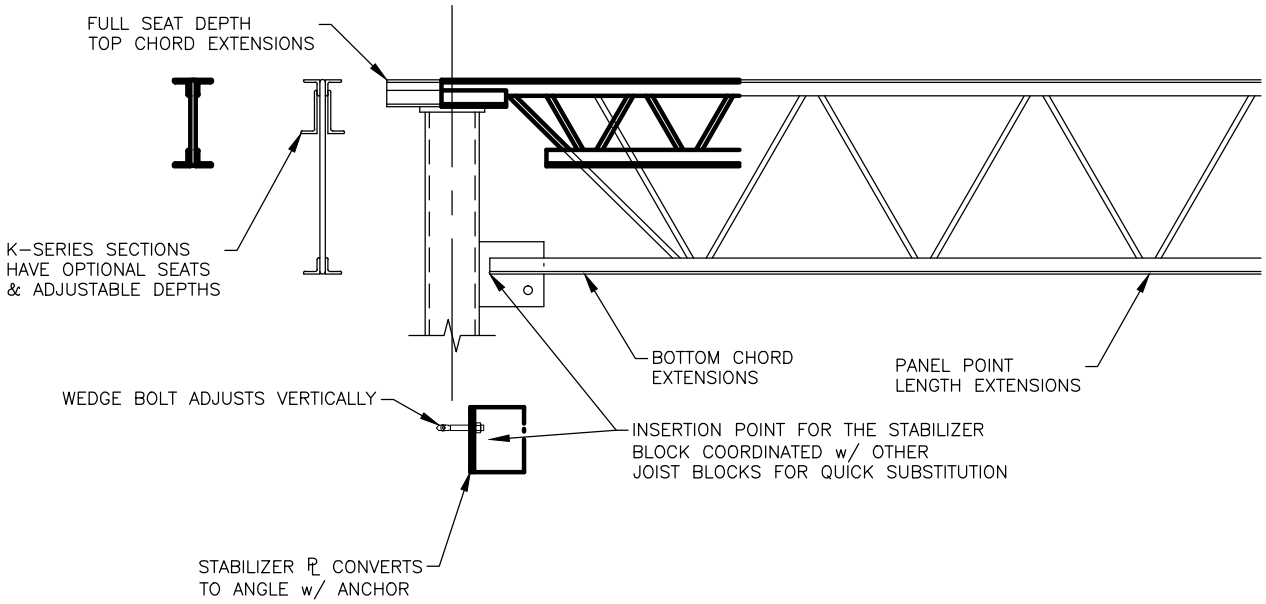
MOMENTCLIPS

INTUICAD SCM GAGE CONNECTORS NOT ONLY PROVIDE QUICK ACCESS TO A LIBRARY OF TYPES BUT ALSO A PLATFORM TO ADD OTHERS THE OFFICE MAY UTILIZE FREQUENTLY.

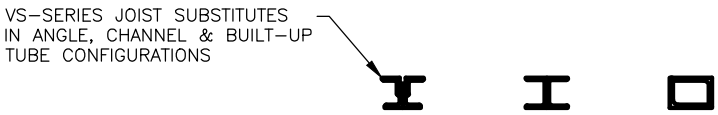
GAGE CONNECTOR OUTLINE LINEWORK COLOR IS BYBLOCK WITH THE REMAINDER OF THE LINEWORK BYLAYER. THE FASTENERS ARE ASSIGNED TO THE "S-ABLT" LAYER WITHIN THE BLOCK.

GAGE CLIP ANGLE BLOCKS ARE INSERTED ON THE "S-WOOD" LAYER.

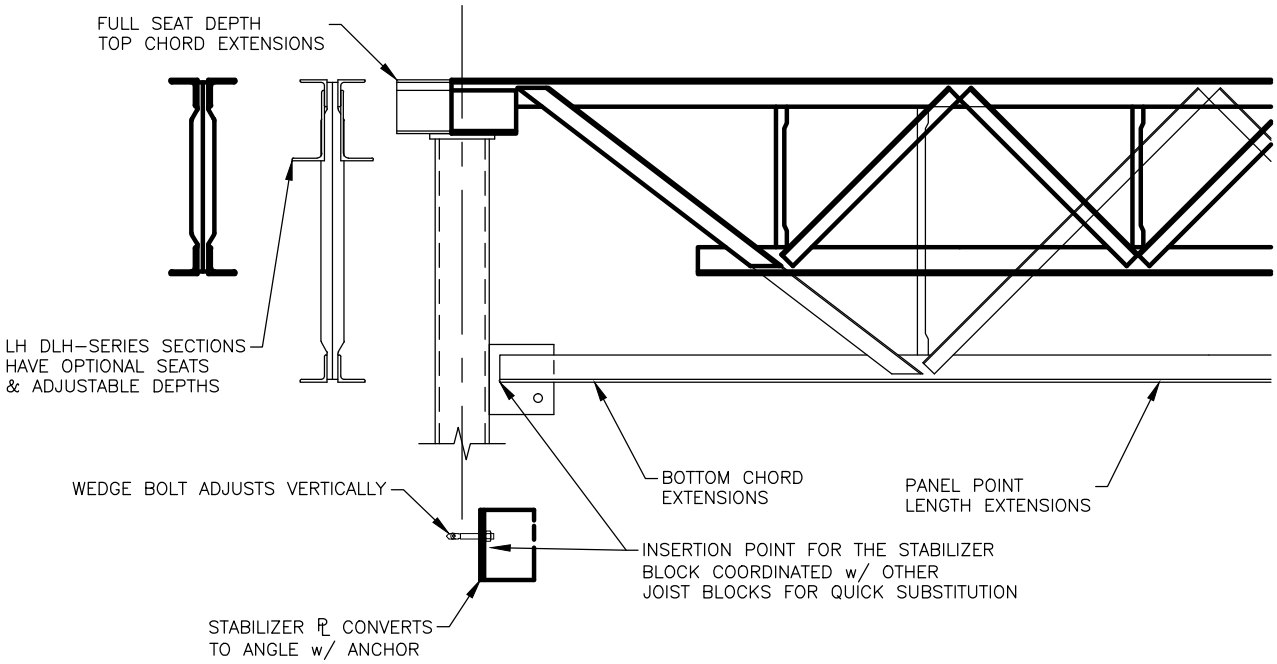
INTUICAD SCM JOISTS ARE SPECIFICALLY ATTUNED TO THE AMBIGUITY INHERENT IN THE MANUFACTURING PROCESS AND THE NEED FOR ACCURATE REPRESENTATION DURING DESIGN. OUR JOIST LIBRARIES HAVE CAREFULLY SELECTED INSERTIONS AND OFFSETS THAT MAKE THEM ONE OF OUR "MOST TIME-SAVING" TOOLS. OUR EDITING CAPABILITIES SIMPLY CANNOT BE MATCHED BY PARAMETRIC ROUTINES.



OUR K-SERIES JOISTS ARE DISTINCTLY DETAILED FOR QUICK RECOGNITION WITH ROUND BAR WEB MEMBERS AND DOUBLE ANGLE CHORDS. BY USING AUTOCAD'S X-CLIP COMMAND TO TRIM, THE DYNAMICS ARE RETAINED (VS EXPLODING) AND WILL PROVIDE SUBSTANTIAL SAVINGS ON FUTURE ADJUSTMENTS. THE FACT THAT THE BLOCK IS TYPICALLY BURIED AMID OTHER LINEWORK DICTATED THE JOIST BE SIMPLY DETAILED AS POSSIBLE.

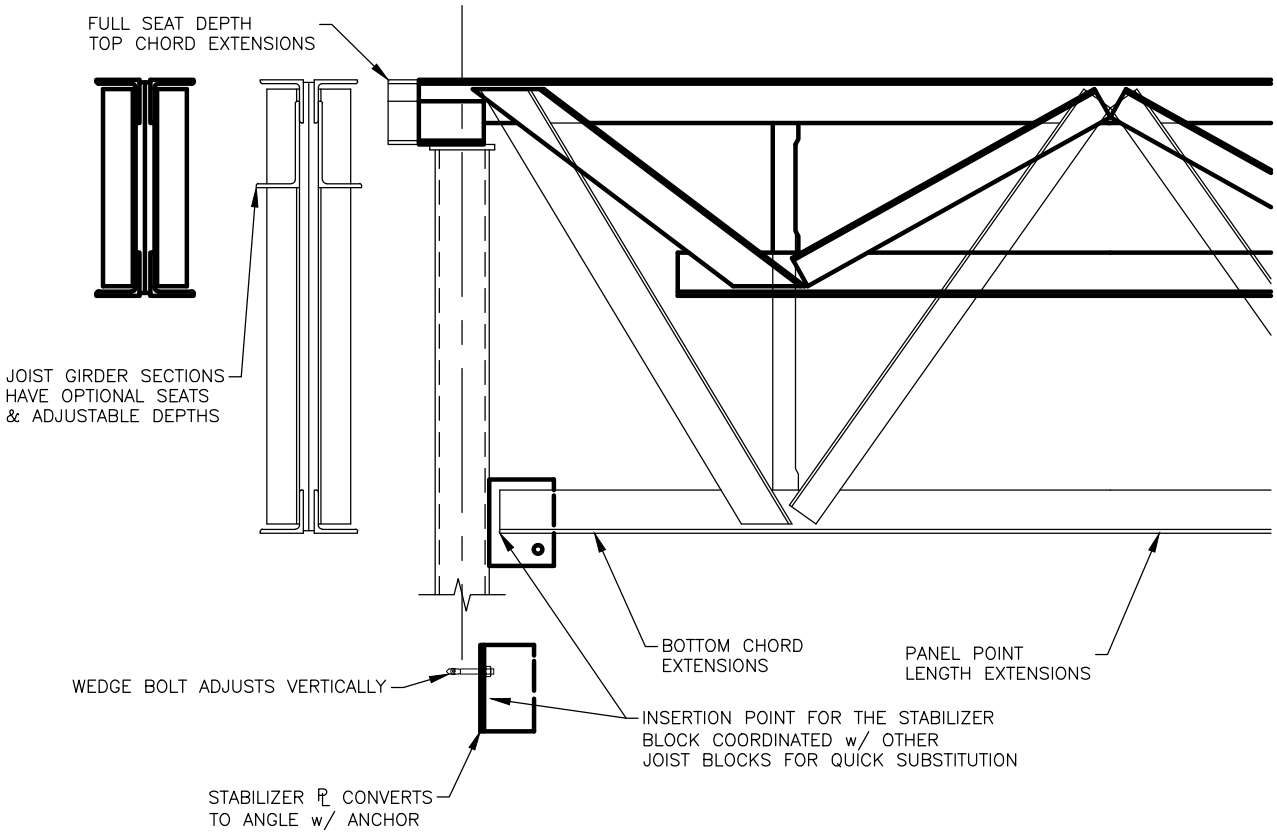


INTUICAD SCM JOISTS ARE SPECIFICALLY ATTUNED TO THE AMBIGUITY INHERENT IN THE MANUFACTURING PROCESS AND THE NEED FOR ACCURATE REPRESENTATION DURING DESIGN. OUR JOIST LIBRARIES HAVE CAREFULLY SELECTED INSERTIONS AND OFFSETS THAT MAKE THEM ONE OF OUR "MOST TIME-SAVING" TOOLS. OUR EDITING CAPABILITIES SIMPLY CANNOT BE MATCHED BY PARAMETRIC ROUTINES.

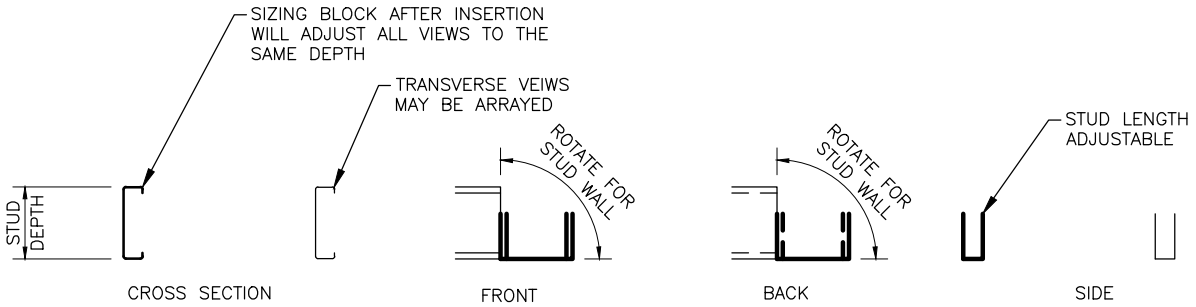


OUR LH & DLH-SERIES JOISTS ARE DISTINCTLY DETAILED FOR QUICK RECOGNITION WITH CRIMPED ANGLE WEB MEMBERS AND DOUBLE ANGLE CHORDS. BY USING AUTOCAD'S X-CLIP COMMAND TO TRIM, THE DYNAMICS ARE RETAINED (VS EXPLODING) AND WILL PROVIDE SUBSTANTIAL SAVINGS ON FUTURE ADJUSTMENTS. THE FACT THAT THE BLOCK IS TYPICALLY BURIED AMID OTHER LINEWORK DICTATED THE JOIST BE SIMPLY DETAILED AS POSSIBLE.

INTUICAD SCM JOISTS ARE SPECIFICALLY ATTUNED TO THE AMBIGUITY INHERENT IN THE MANUFACTURING PROCESS AND THE NEED FOR ACCURATE REPRESENTATION DURING DESIGN. OUR JOIST LIBRARIES HAVE CAREFULLY SELECTED INSERTIONS AND OFFSETS THAT MAKE THEM ONE OF OUR "MOST TIME-SAVING" TOOLS. OUR EDITING CAPABILITIES SIMPLY CANNOT BE MATCHED BY PARAMETRIC ROUTINES.

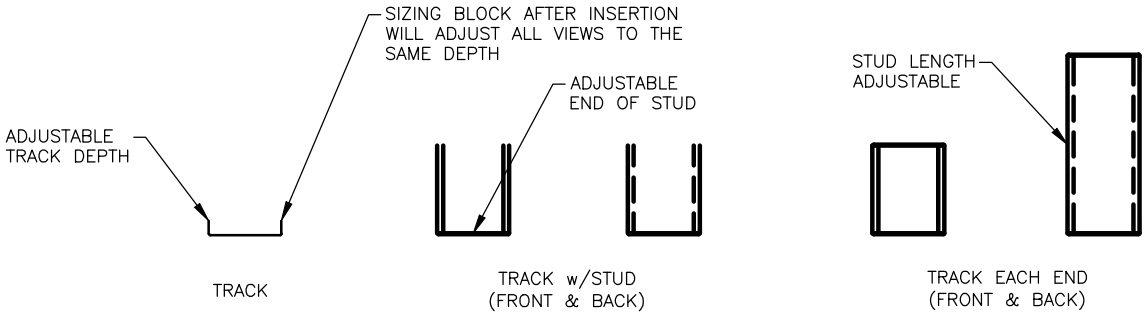


OUR JOIST GIRDERS ARE DISTINCTLY DETAILED FOR QUICK RECOGNITION WITH CRIMPED ANGLE WEB MEMBERS AND DOUBLE ANGLE CHORDS. BY USING AUTOCAD'S X-CLIP COMMAND TO TRIM, THE DYNAMICS ARE RETAINED (VS EXPLODING) AND WILL PROVIDE SUBSTANTIAL SAVINGS ON FUTURE ADJUSTMENTS. THE FACT THAT THE BLOCK IS TYPICALLY BURIED AMID OTHER LINEWORK DICTATED THE JOIST BE SIMPLY DETAILED AS POSSIBLE.



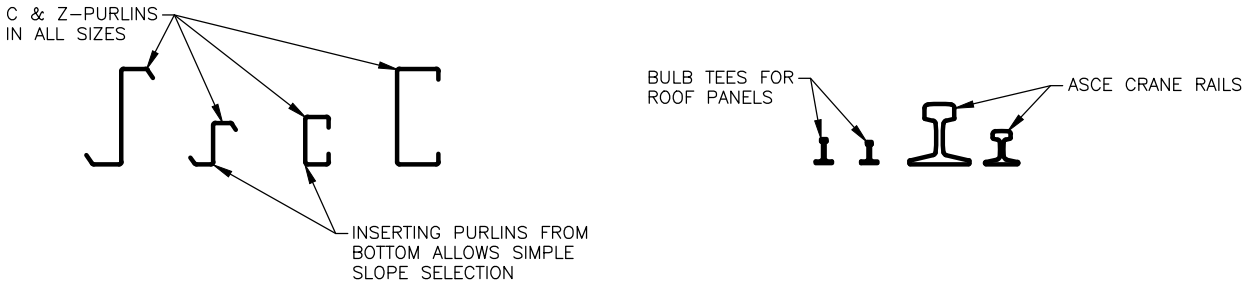
THE STUD/JOIST BLOCK IS A MODEL OF COORDINATED DYNAMICS. DRAFTERS MAY INSERT IT ONLY ONCE PER DRAWING BECAUSE IT OFTEN TAKES FEWER "CLICKS" TO TURN ONE INTO ANOTHER THAN TO RE-INSERT FROM THE MENU OR TOOLBAR.

THE STUD/JOIST BLOCK CUT EDGE LINEWORK COLOR IS BYBLOCK. THE REMAINDER IS BYLAYER. THE HIDDEN LINE LINETYPE IS BYBLOCK.

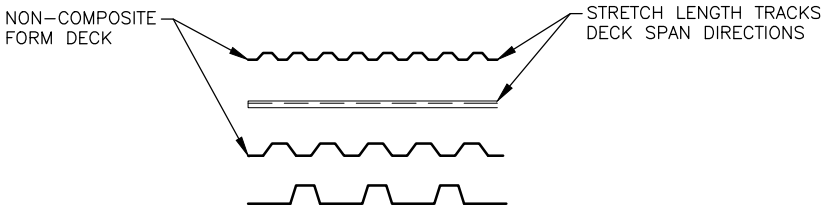


THE TRACK/WALL BLOCK PROVIDES AN EXCELLENT COMPLIMENT TO THE STUD/JOIST BLOCK. THE INSERTION POINTS AND GRIP LOCATIONS HAVE BEEN CAREFULLY COORDINATED BETWEEN THEM.

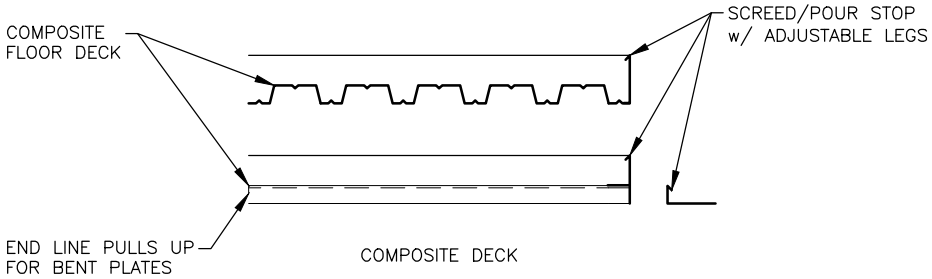
THE TRACK/WALL BLOCK CUT EDGE LINEWORK COLOR IS BYBLOCK. THE REMAINDER IS BYLAYER. THE HIDDEN LINE LINETYPE IS BYBLOCK.



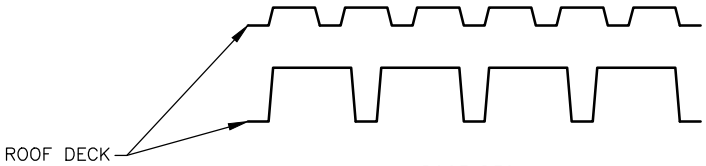
THE C-PURLIN, Z-PURLIN, BULB TEE AND CRAIN RAIL BLOCKS ALL HAVE LINEWORK BYLAYER.



NON-COMPOSITE FORM



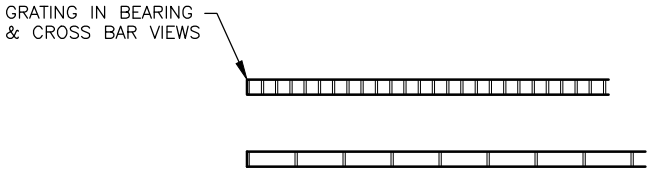
COMPOSITE DECK



ROOF DECK

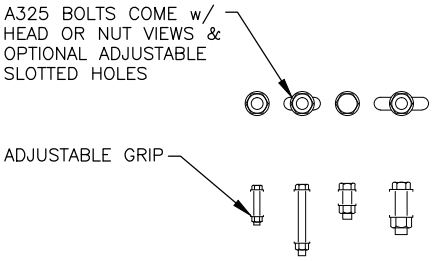
THE METAL FORM/DECK BLOCK PROFILES HAVE BEEN CAREFULLY WEIGHED FOR DRAFTING COHESION AS WELL AS INDUSTRY ACCURACY. WHERE ALL OTHER THINGS WERE EQUAL, PREFERENCE WAS GIVEN TO THOSE PROFILES WITH EQUAL PITCH SINCE THIS ALLOWS INTERCHANGABILITY AMONG TYPES.

ALL METAL FORM/DECK BLOCK CUT EDGE LINEWORK IS BYBLOCK. ALL BLOCKS ARE INSERTED ON THE "S-DECK" LAYER.

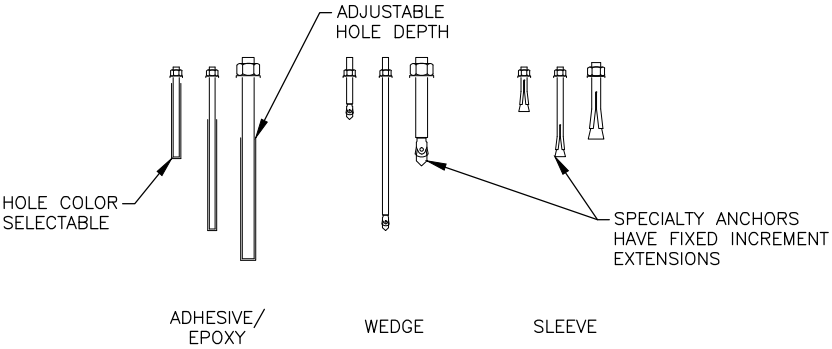


THE GRATING BLOCK PROFILES HAVE BEEN CAREFULLY WEIGHED FOR DRAFTING COHESION AS WELL AS INDUSTRY ACCURACY. DIFFERENTIAL BAR SPACING FOR DEEPER BARS WAS IGNORED IN ORDER TO ALLOW A SINGLE OPERATION TO ADJUST THE GRATING DEPTH.

ALL GRATING BLOCK CUT EDGE LINEWORK IS BYBLOCK. THE BLOCT IS INSERTED ON THE "S-DECK" LAYER.

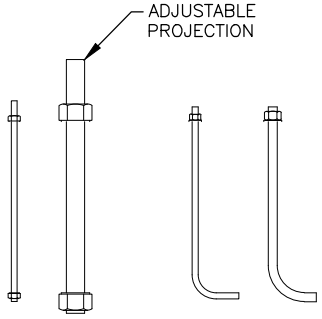


THE A325 BOLT BLOCKS ARE INSERTED ON THE "S-ABLT" LAYER WITH ALL BLOCK LINEWORK COLOR BYLAYER.



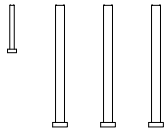
THE ADHESIVE WEDGE AND SLEEVE ANCHOR BLOCKS ARE INSERTED ON THE "S-ABLT" LAYER WITH ALL BLOCK LINEWORK COLOR BYLAYER.

THE WEDGE AND SLEEVE ANCHORS ARE MODELED AFTER INDUSTRY FAVORITES.



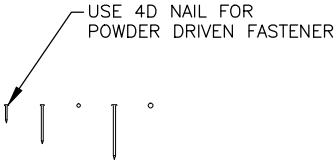
THE ANCHOR BOLT BLOCKS ARE INSERTED ON THE "S-ABLT" LAYER WITH ALL BLOCK LINEWORK COLOR BYLAYER.

THE ANCHOR BOLTS ARE DRAWN TO ASTM SPECIFICATIONS.

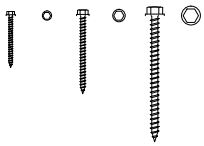


THE HEADED STUD BLOCKS ARE INSERTED ON THE "S-ABLT" LAYER WITH ALL BLOCK LINEWORK COLOR BYLAYER.

INDIVIDUAL HEADED STUD SIZES WITHIN THE BLOCK CAN BE MADE TO STRETCH AT DIFFERENT INCREMENTS OR IN UNISON. THE ARRAY CAN ALSO BE SET TO INDIVIDUAL STUD SIZES.

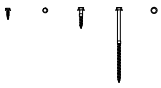


THE NAIL BLOCKS ARE INSERTED ON THE "S-ABLT" LAYER WITH ALL BLOCK LINEWORK COLOR BYLAYER.



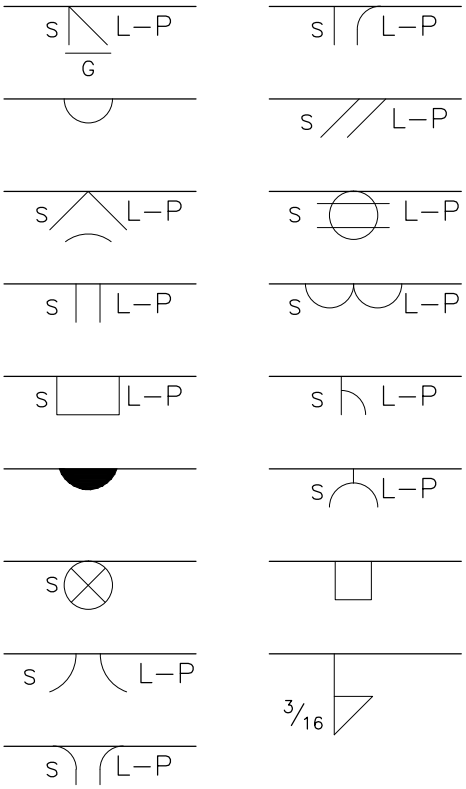
THE LAG SCREW BLOCK ELEVATION VIEWS STRETCH IN THREAD PITCH INCREMENTS.

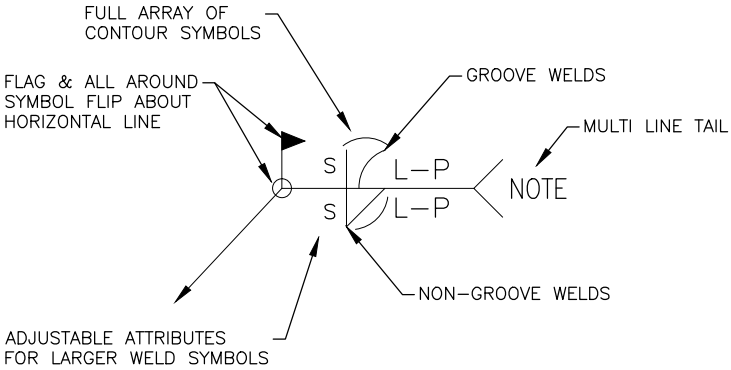
THE LAG SCREW BLOCK IS INSERTED ON THE "S-ABLT" LAYER WITH ALL BLOCK LINEWORK COLOR BYLAYER.



THE TEK/SDS BLOCK ELEVATION VIEWS DO NOT HAVE STRETCH DYNAMICS SINCE THEY HAVE FIXED THREAD LENGTHS.

THE TEK/SDS BLOCK IS INSERTED ON THE "S-ABLT" LAYER WITH ALL BLOCK LINEWORK COLOR BYLAYER.

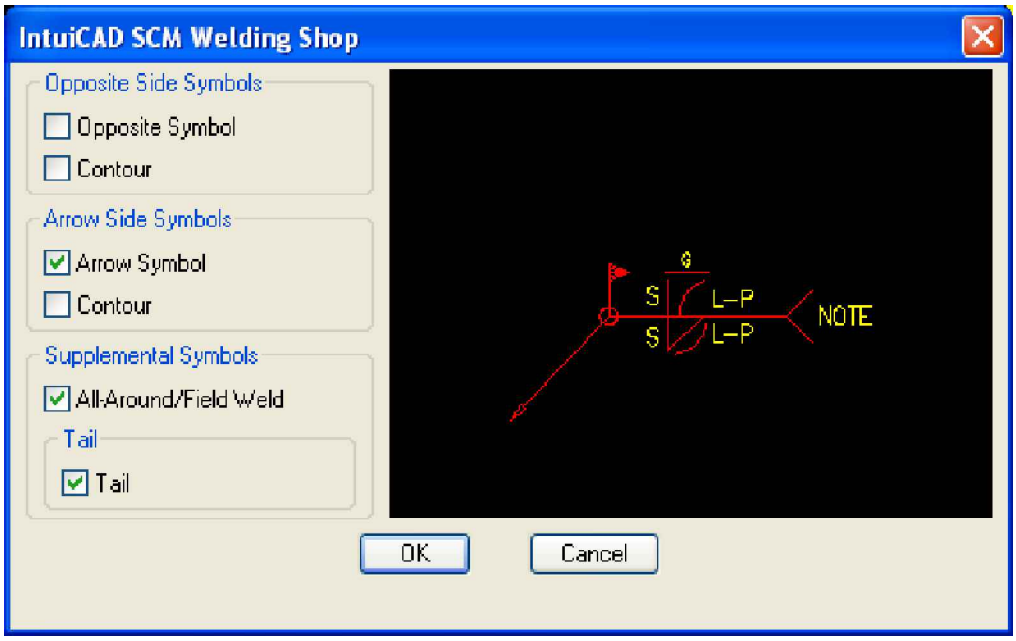


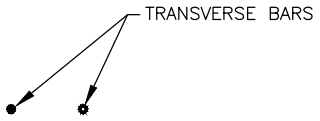


WELD SYMBOL IS GRIP REVERSABLE:

1. GRIP AND FLIP SUPPLEMENTAL SYMBOLS.
2. GRIP AND FLIP TAIL AND DRAG TEXT.
3. GRIP AND DRAG LEADER TO OPPOSITE END.

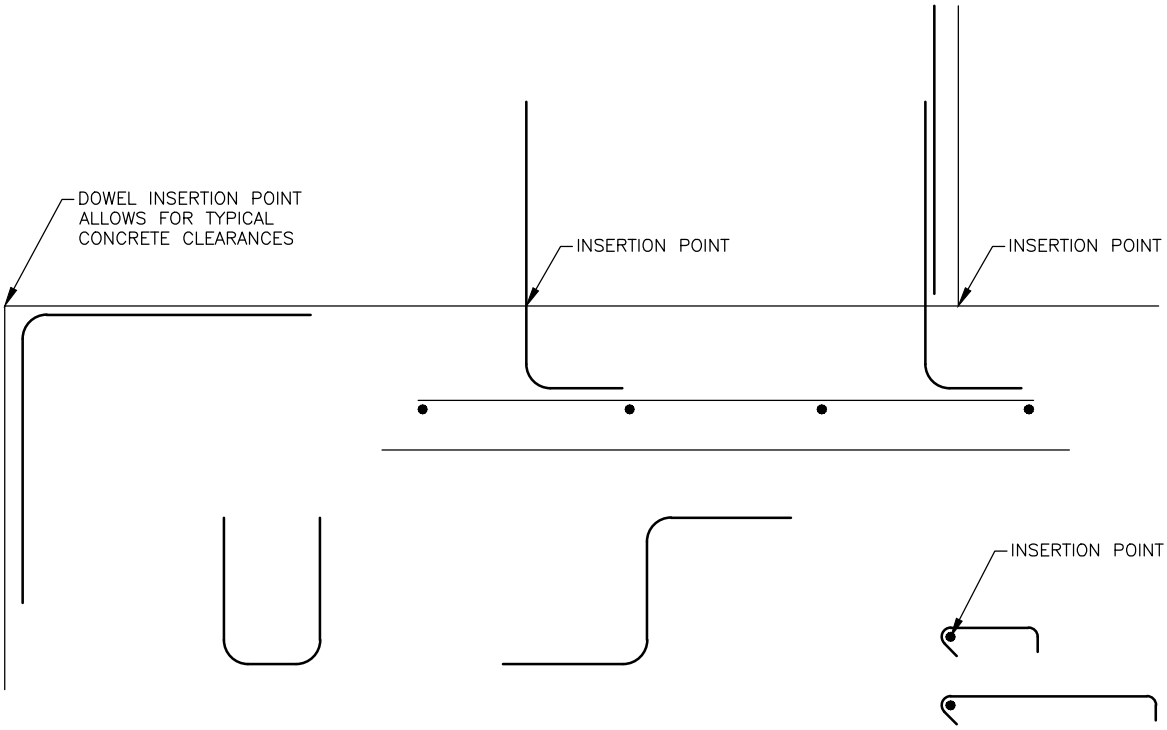
NO OBJECTS BETWEEN THE ENDPOINTS OF THE HORIZONTAL LINE ARE EVER FLIPPED L TO R.





TRANSVERSE REINFORCING BARS ARE MADE USING AUTOCAD'S DONUT COMMAND.

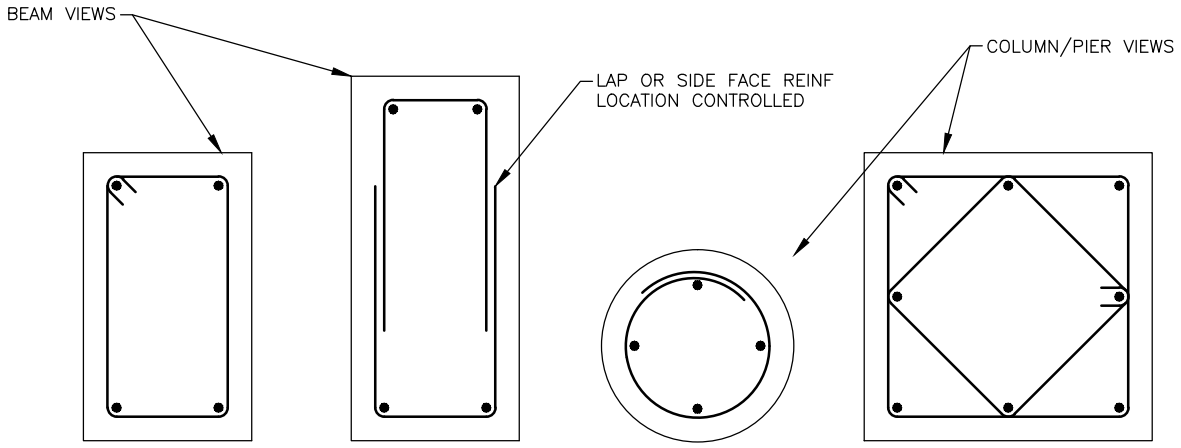
THE SOLID REINFORCING BAR HAS AN OUTER DIAMETER OF ¾" AND AN INNER DIAMETER OF 0. THE HOLLOW BAR HAS AN OUTSIDE DIAMETER OF ¾" AND AN INNER DIAMETER OF ⅜"



THE MAGIG NUMBER TO REMEMBER TO MATCH THE LINEWORK IN THE REINFORCING DOWEL BLOCK IS ¾". ALL CLEARANCES BETWEEN BARS AS WELL AS STIRRUP/TIE RADII ARE ¾". DOWEL BAR BEND RADII ARE DRAWN AT 2". WHILE NEITHER OF THESE NUMBERS REPRESENTS ACTUAL FIELD CONDITIONS, THEY FUNCTION WELL IN LEAVING NOTICEABLE GAPS BETWEEN THE LINEWORK AND MAKE LAPPING LESS DISTORTED.

ALTHOUGH THE LINEWORK IS QUITE SIMPLISTIC, THE DOWEL BLOCK HANDLES SECONDARY EDITING FAR MORE EFFICIENTLY THAN THE EXPLODED LINEWORK.

THE REINFORCING DOWEL BLOCK LINEWORK ASSUMES ZERO BAR WIDTH FOR THE PURPOSES OF MEASURING FOR CLEARANCES.



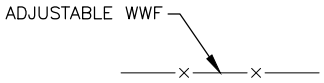
THE MAGIG NUMBER TO REMEMBER TO MATCH THE LINEWORK IN THE STIRRUP/TIE BLOCK IS ¾". ALL CLEARANCES BETWEEN BARS AS WELL AS STIRRUP/TIE RADII ARE ¾". WHILE THIS NUMBER MAY NOT REPRESENT ACTUAL FIELD CONDITIONS, IT FUNCTIONS WELL IN LEAVING NOTICEABLE GAPS BETWEEN THE LINEWORK AND MAKE LAPPING LESS DISTORTED.

ALTHOUGH THE LINEWORK IS QUITE SIMPLISTIC, THE BLOCK IS EASILY EXPANDED INTO PIER, COLUMN AND GRADE BEAM DETAILS.

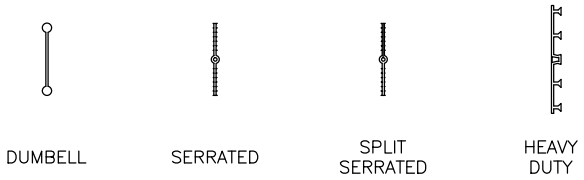
THE REINFORCING STIRRUP/TIE BLOCK LINEWORK ASSUMES ZERO BAR WIDTH FOR THE PURPOSES OF MEASURING FOR CLEARANCES.

THE THREADED BAR COUPLER PROVIDES A QUICK GENERIC SOLUTION FOR DELINEATING FIELD SPLICES.

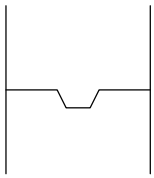
THE THREADED BAR COUPLER BLOCK LINEWORK COLOR IS BYBLOCK. IT IS INSERTED ON THE "S-RBAR" LAYER.



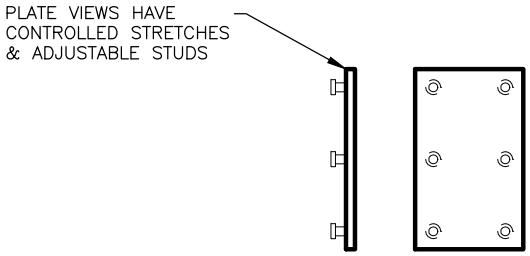
THE WELDED WIRE FABRIC BLOCK ALLOWS FAST ACCESS TO WIRE MESH OR FENCING.



THE WATERSTOP BLOCK LINEWORK IS BYBLOCK. IT IS INSERTED ON THE "S-RBAR" LAYER.

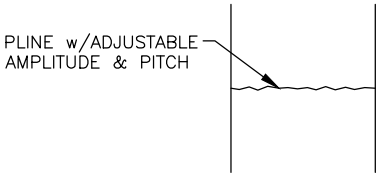


THE SHEAR KEY TOOL IS A PARAMETRIC ROUTINE THAT CREATES A CENTERED 2x4 KEY IN A CONCRETE WALL. THE KEY IS DRAWN ON THE CURRENT LAYER.

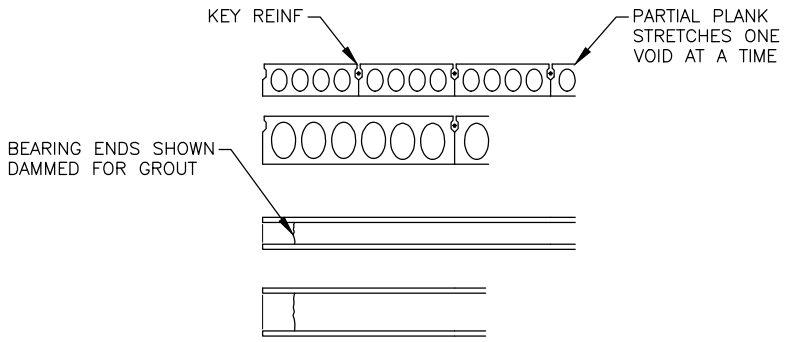


THE EMBEDDED PLATE BLOCK LINEWORK COLOR IS BYLAYER WITH THE BOLTHEADS ASSIGNED. THE BLOCK IS ASSIGNED A HIDDEN LINETYPE.

DO NOT PUT DYNAMIC FASTENERS INSIDE OF EMBEDDED PLATE BLOCK. DOING SO MAY CAUSE THE DYNAMICS OF THE ENTIRE BLOCK TO FAIL AS WELL AS THE DYNAMICS IN OTHER OCCURANCES OF THE FASTENER BLOCK.



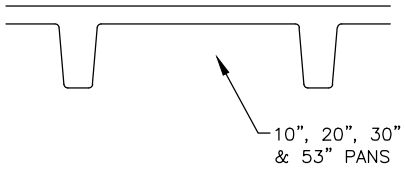
THE ROUGHEN TOOL IS A PARAMETRIC ROUTINE THAT CREATES A JAGGED POLYLINE. THE POLYLINE CHARACTERISTICS MATCH THOSE OF THE SELECTED LINE.



THE HOLLOW CORE PRECAST PLANK BLOCK PROFILES HAVE BEEN CAREFULLY WEIGHED FOR DRAFTING COHESION AS WELL AS INDUSTRY ACCURACY. WHERE ALL OTHER THINGS WERE EQUAL, PREFERENCE WAS GIVEN TO THOSE PROFILES WITH EQUAL PITCH SINCE THIS ALLOWS INTERCHANGABILITY AMONG PLANK DEPTHS.

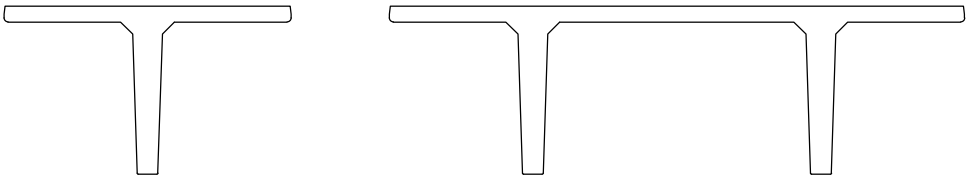
NEARLY ALL HOLLOW CORE PRECAST PLANK BLOCK EDGE LINEWORK IS BYLAYER. THE EXCEPTIONS ARE THE END LINE IN ELEVATION (BYBLOCK) AND THE REINFORCING BARS (ASSIGNED WITHIN THE BLOCK).

THE HOLLOW CORE PRECAST PLANK BLOCK IS INSERTED ON THE "S-SECT-CONC" LAYER.



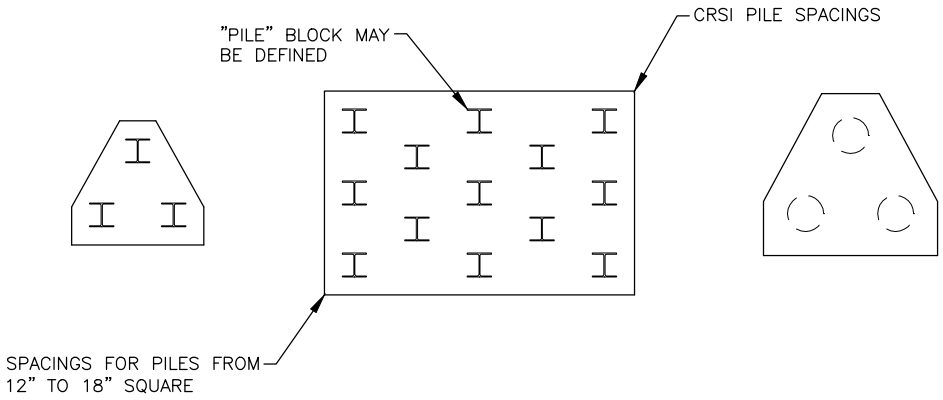
THE CONCRETE PAN JOIST BLOCK PROVIDES QUICK ACCESS TO WIDE SPAN JOIST PROFILES.

THE CONCRETE PAN JOIST BLOCK IS INSERTED ON THE "S-SECT-CONC" LAYER.



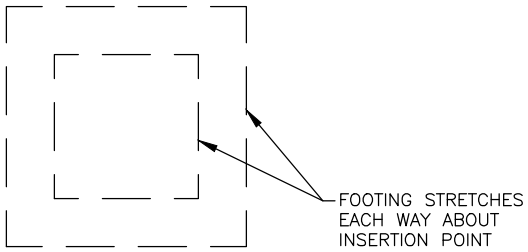
THE SINGLE AND DOUBLE TEE BLOCKS PROVIDES QUICK ACCESS TO TEE SECTION PROFILES.

THE SINGLE AND DOUBLE TEE BLOCKS ARE INSERTED ON THE "S-SECT-CONC" LAYER.

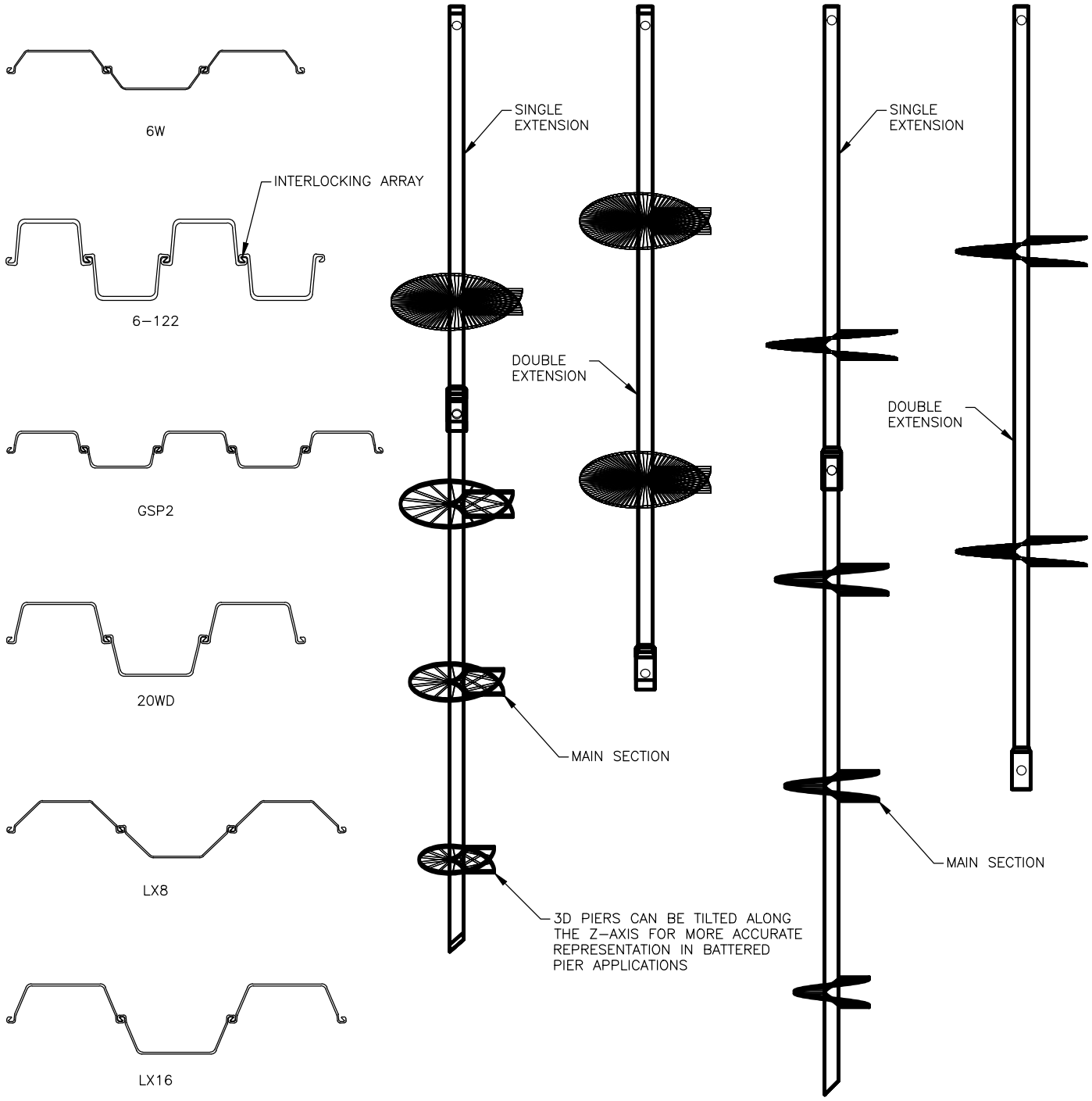


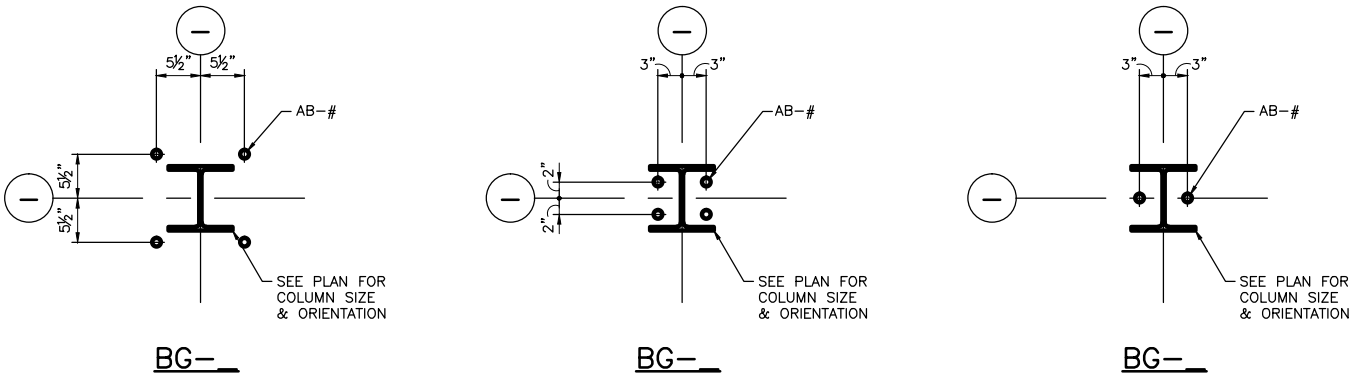
THE CONCRETE PILE CAP BLOCKS ARE EMBEDDED WITH PILE BLOCKS DENOTING THEIR SIZE FAMILY (IN INCHES). ONCE THE PILE CAPACITY HAS BEEN SELECTED, REDEFINING THE PILE BLOCK WITH THE APPROPRIATE CROSS SECTION WILL UPDATE THE BLOCKS OF THAT FAMILY OF CAPS.

A BLOCK OF DIMENSIONED CAP DETAILS IS INCLUDED IN THE BONUS TOOLS.

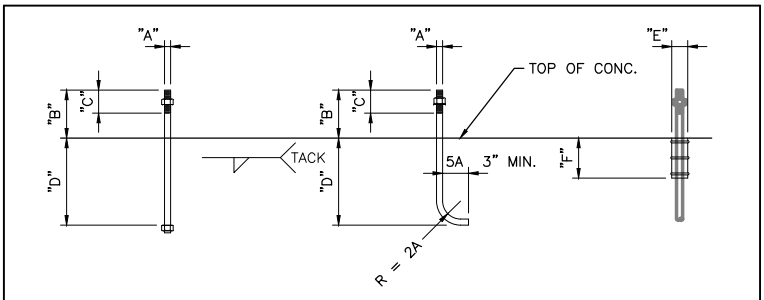


THE FOOTING BLOCK MAKES A GREAT PLACEHOLDER IN THE EARLY STAGES OF FOUNDATION DESIGN. SIMPLE BUT HIGHLY EFFECTIVE.


6W
INTERLOCKING ARRAY
6–122
GSP2
20WD
LX8
LX16
SINGLE EXTENSION
DOUBLE EXTENSION
MAIN SECTION
3D PIERS CAN BE TILTED ALONG THE Z–AXIS FOR MORE ACCURATE REPRESENTATION IN BATTERED PIER APPLICATIONS
SINGLE EXTENSION
DOUBLE EXTENSION
MAIN SECTION



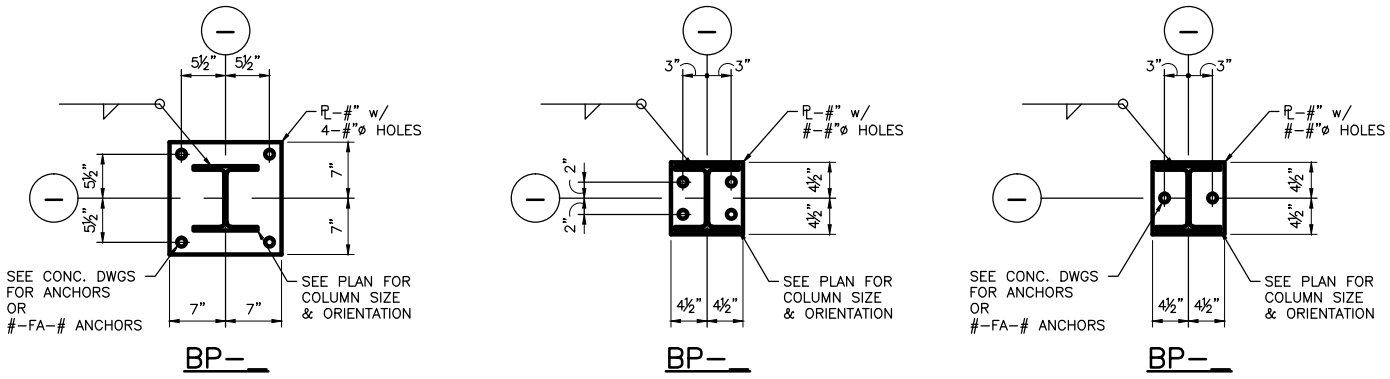
THE BOLT GAGE BLOCK WAS AN EXERCISE TO DIVIDE INFORMATION REQUIRED BY THE CONCRETE CONTRACTOR FROM THAT REQUIRED BY THE STEEL FABRICATOR. ANCHOR BOLT BOLT GAGES WERE INTEGRATED WITH BASE PLATE DETAILS IN A WAY THAT MADE IT EASIER MATCH THE TWO EVEN WHEN THE PACKAGES WERE BID AT DIFFERENT TIMES



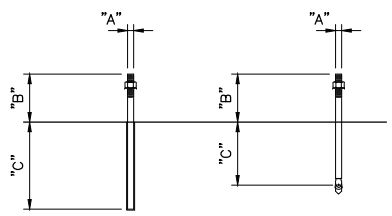
MARK	NO REQ'D	TYPE	DIM "A"	DIM "B"	DIM "C"	DIM "D"	SLEEVE		REMARKS
							DIM "E"	DIM "F"	
–	–	–	–	–	–	–	–	–	–

ALL ANCHOR BOLTS TO BE FURNISHED WITH HEAVY HEX NUTS AND HARDENED WASHER.

IT SEEMED ONLY NATURAL TO INCLUDE THE ACCOMPANYING SCHEDULE AS THE LAST VISIBILITY.



THE BASE PLATE BLOCK WAS AN EXERCISE TO DIVIDE INFORMATION REQUIRED BY THE CONCRETE CONTRACTOR FROM THAT REQUIRED BY THE STEEL FABRICATOR. ANCHOR BOLT BOLT GAGES WERE INTEGRATED WITH BASE PLATE DETAILS IN A WAY THAT MADE IT EASIER MATCH THE TWO EVEN WHEN THE PACKAGES WERE BID AT DIFFERENT TIMES

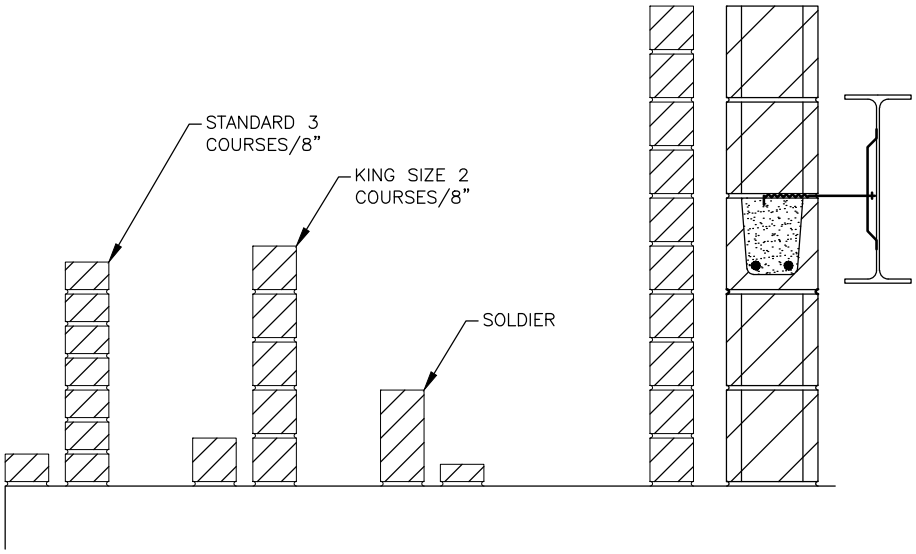


TYPE–I (ADHESIVE) TYPE II (MECHANICAL)

MARK	NO REQ'D	DIM "A"	DIM "B"	DIM "C"	REMARKS
–	–	–	–	–	–

SEE GENERAL NOTES FOR ANCHOR SPECIFICATIONS.

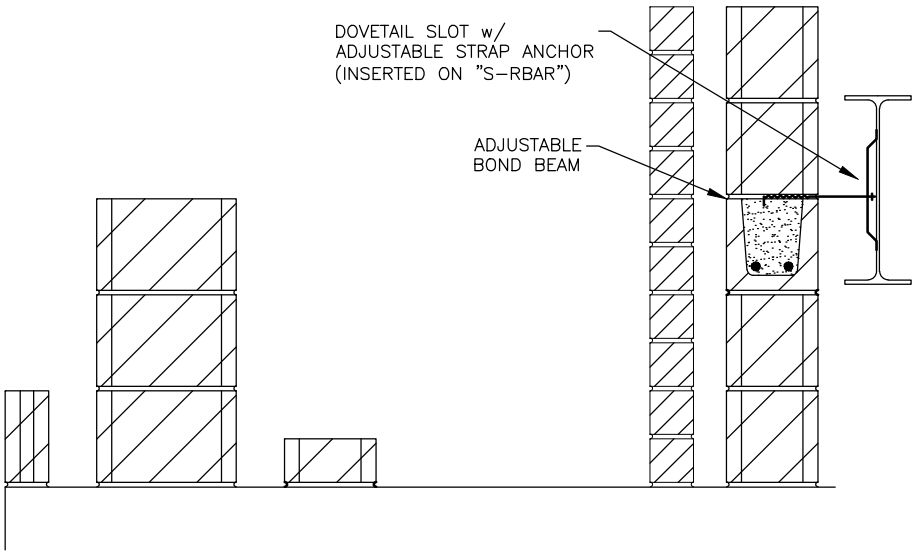
IT SEEMED ONLY NATURAL TO INCLUDE THE ACCOMPANYING SCHEDULE AS THE LAST VISIBILITY.



THE BRICK UNIT BLOCK IS VERSITILE AND HIGHLY FUNCTIONAL WAY TO ENSURE THE ALIGNMENT OF JOINTS.

THE BRICK UNIT BLOCK EDGE LINEWORK IS BYLAYER. THE HATCHING IS ASSIGNED WITHIN THE BLOCK.

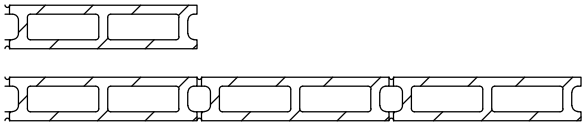
THE BRICK UNIT BLOCK IS INSERTED ON THE "S-BRIK" LAYER.



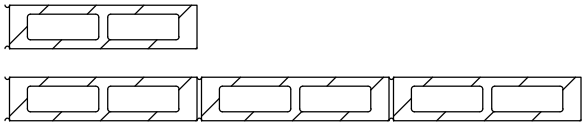
THE CMU BLOCK IS VERSITILE AND HIGHLY FUNCTIONAL WAY TO ENSURE THE ALIGNMENT OF JOINTS.

THE CMU BLOCK EDGE LINEWORK IS BYLAYER. THE HATCHING IS ASSIGNED WITHIN THE BLOCK. BOND BEAM BLOCK REBAR IS ASSIGNED WITHIN THE BLOCK.

THE CMU BLOCK IS INSERTED ON THE "S-CMUW" LAYER.



REGULAR

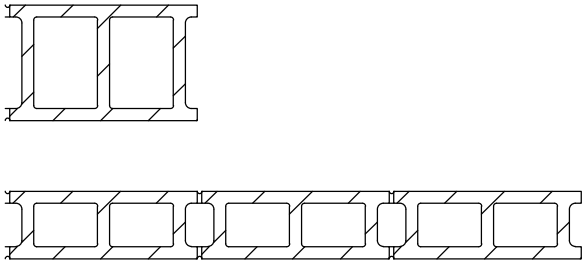


PLAIN

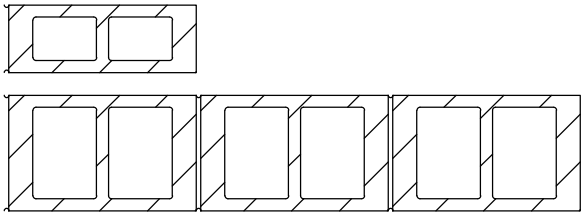
THE 4" CMU PLAN BLOCK IS VERSITILE AND HIGHLY FUNCTIONAL WAY TO ENSURE THE ALIGNMENT OF JOINTS. THE WALLS ARE THINNER THAN THE 6"-12" CMU PLAN BLOCK.

THE 4" CMU PLAN BLOCK EDGE LINEWORK IS BYLAYER. THE HATCHING IS ASSIGNED WITHIN THE BLOCK.

THE 4" CMU PLAN BLOCK IS INSERTED ON THE "S-CMUW" LAYER.



REGULAR

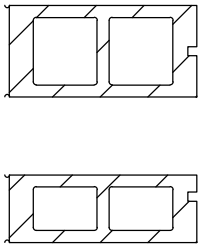


PLAIN

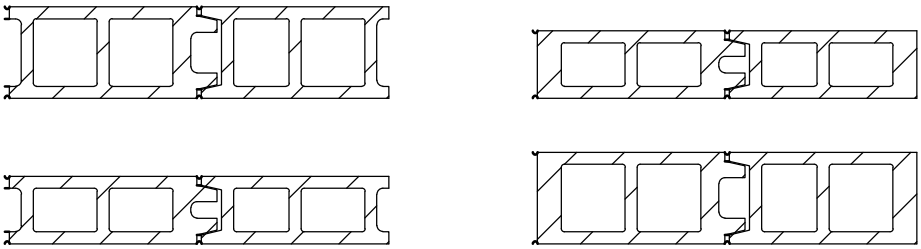
THE 6"-12" CMU PLAN BLOCK IS VERSITILE AND HIGHLY FUNCTIONAL WAY TO ENSURE THE ALIGNMENT OF JOINTS. THE WALLS ARE THICKER THAN THE 4" CMU PLAN BLOCK.

THE 6"-12" CMU PLAN BLOCK EDGE LINEWORK IS BYLAYER. THE HATCHING IS ASSIGNED WITHIN THE BLOCK.

THE 6"-12" CMU PLAN BLOCK IS INSERTED ON THE "S-CMUW" LAYER.

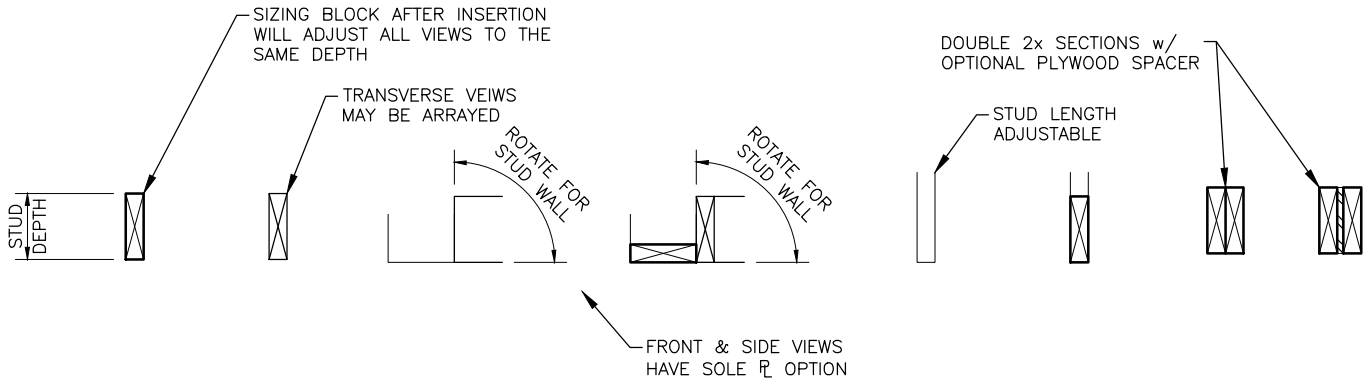


THE 6"-12" CMU PLAN BLOCK ALSO CONTAINS AN ADJUSTABLE SASH BLOCK.



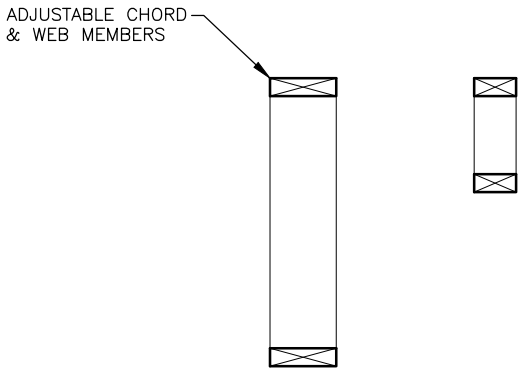
REGULAR PLAIN

THE 6"-12" CMU PLAN BLOCK ALSO CONTAINS EXPANSION JOINT BLOCK PAIRS.

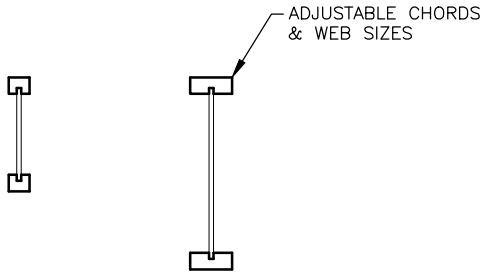


THE WOOD STUD BLOCK IS A MODEL OF COORDINATED DYNAMICS. DRAFTERS MAY INSERT IT ONLY ONCE PER DRAWING BECAUSE IT OFTEN TAKES FEWER "CLICKS" TO TURN ONE INTO ANOTHER THAN TO RE-INSERT FROM THE MENU OR TOOLBAR.

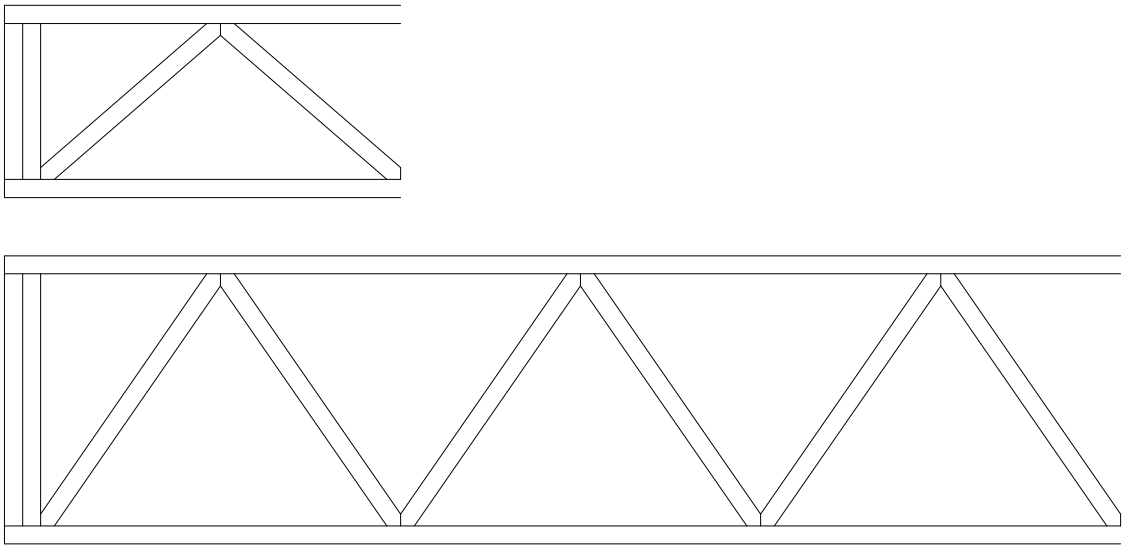
THE STUD/JOIST BLOCK CUT EDGE LINEWORK COLOR IS BYBLOCK. THE REMAINDER IS BYLAYER.



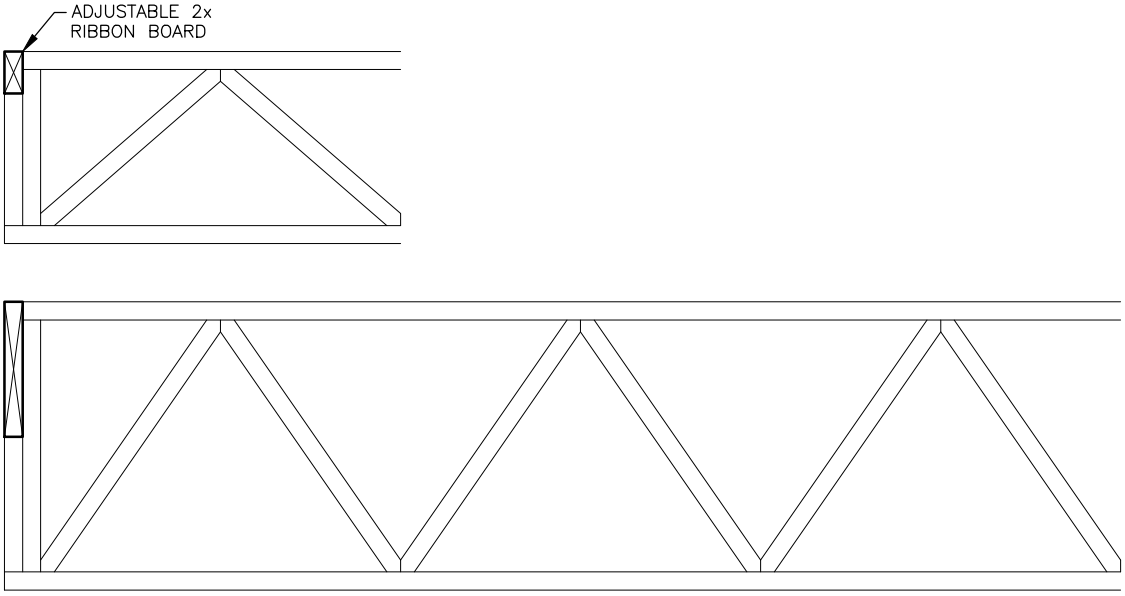
THE WOOD TRUSS SECTION BLOCK CUT EDGE LINEWORK COLOR IS BYBLOCK. THE REMAINDER IS BYLAYER.



THE TRUSS JOIST SECTION BLOCK CUT EDGE LINEWORK COLOR IS BYBLOCK. THE REMAINDER IS BYLAYER.

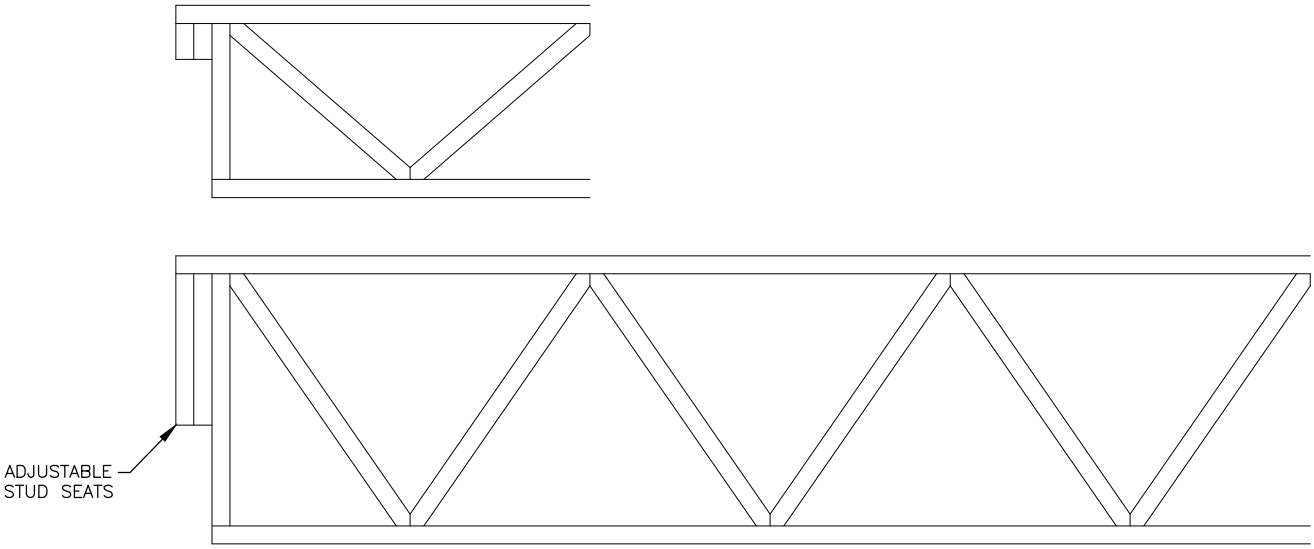


RIM



RIBBON

THE BOTTOM CHORD BEARING TRUSS BLOCK EDGE LINEWORK COLOR IS BYBLOCK. THE REMAINDER IS BYLAYER.



STUD SEAT

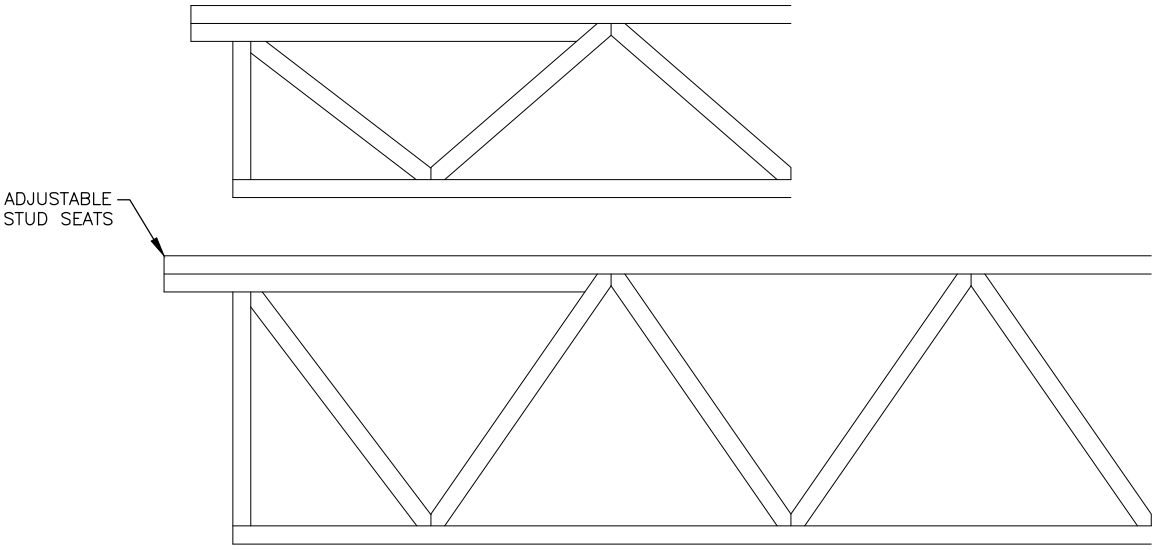
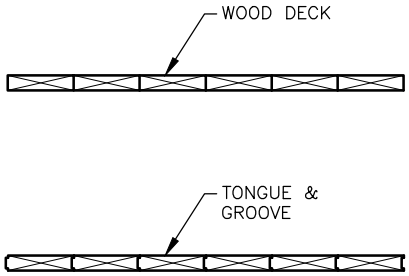
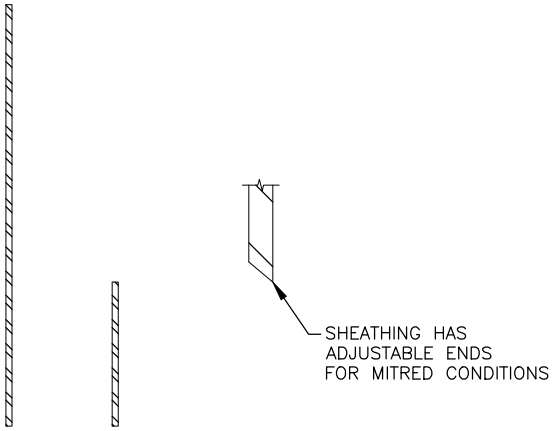


PLATE SEAT

THE TOP CHORD BEARING TRUSS BLOCK LINEWORK COLOR IS BYLAYER.

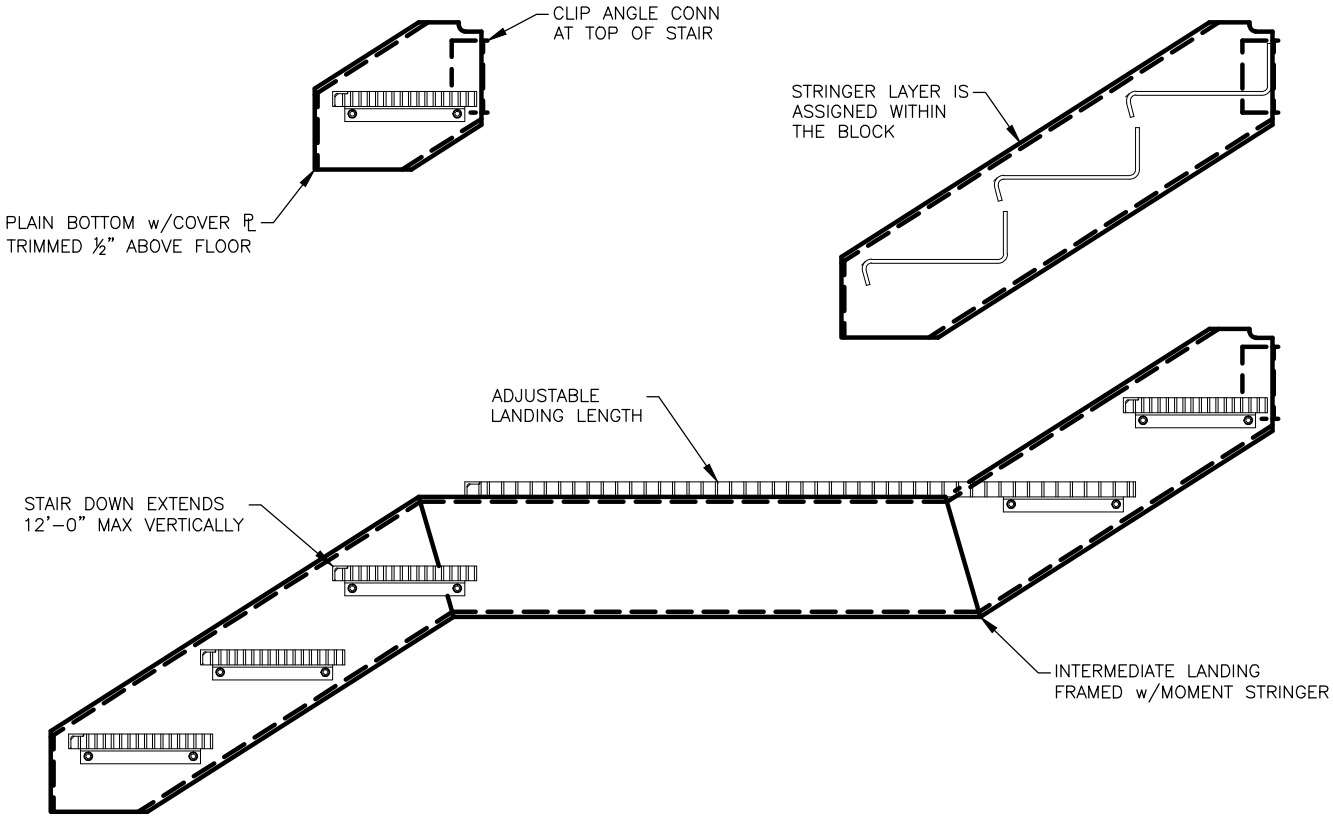


THE WOOD DECK BLOCK CUT EDGE LINEWORK COLOR IS BYBLOCK. THE REMAINDER IS BYLAYER.



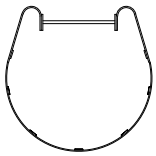
THE SHEATHING BLOCK LINEWORK COLOR IS BYLAYER. THE HATCHING IS ASSIGNED WITHIN THE BLOCK.

THE STAIR BLOCK STARTS WITH A VARIETY OF TYPICAL AND NOT SO TYPICAL TREADS.



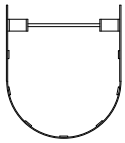
STAIR ELEVATIONS MAY BE ONE OF THE MOST TEDIOUS DRAWINGS TO CREATE. ONCE A STAIR HAS BEEN CREATED, ANY CHANGES TO THE STRUCTURAL DESIGN OF THE STAIR PUTS A MAJOR WORKLOAD DIRECTLY BACK ON THE DRAFTER... POTENTIALLY HUNDREDS OF LINES THAT NEED TO BE RECREATED; POTENTIALLY HOURS OF ADDITIONAL LABOR. INTUICAD'S STAIR DYNAMICS SOLVE THIS PROBLEM IN GRAND STYLE! OUR DYNAMIC STAIR HAS BEEN CREATED IN ITS ENTIRETY WITH PLATFORMS AND LANDINGS INCLUDED. OUR STAIR DEFAULTS TO THE STANDARD 7" RISE, 11" TREAD, AND 1" OVERHANG, BUT EVEN THESE CAN BE ADJUSTED TO SUIT OTHER CONFIGURATIONS.

BAR MIDDLE



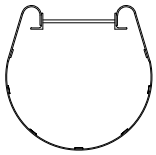
BAR MIDDLE CAGE

BAR TOP

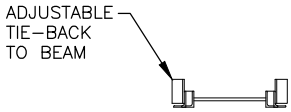


BAR TOP CAGE

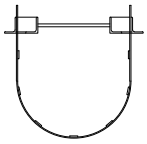
ANGLE MIDDLE



ANGLE MIDDLE CAGE

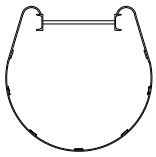


ANGLE TOP



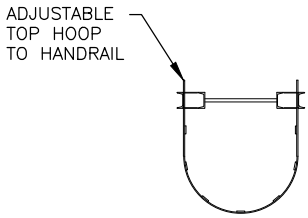
ANGLE TOP CAGE

CHANNEL MIDDLE

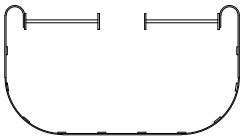


CHANNEL MIDDLE CAGE

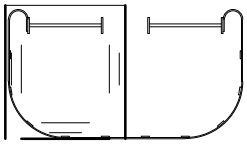
CHANNEL TOP



CHANNEL TOP CAGE



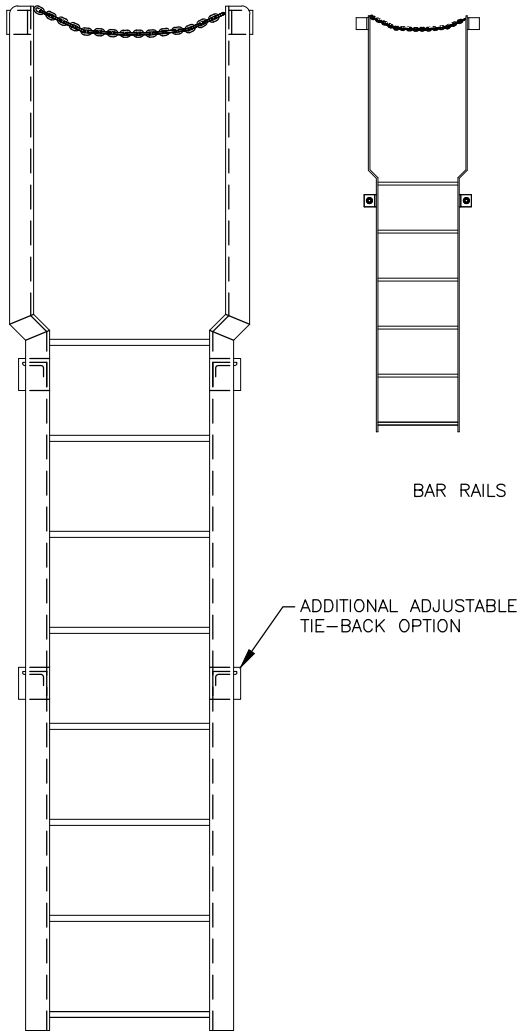
SIDESTEP CAGE

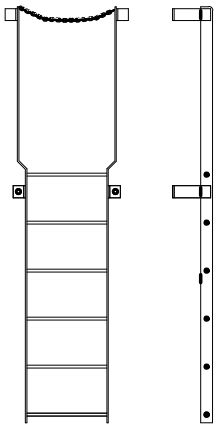


SIDESTEP CAGE w/FRAMING

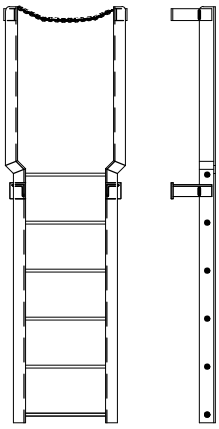
THE PLAN LADDER BLOCK CAN INTERCHANGE BETWEEN ALL OF THE VIEWS ABOVE. THE BLOCK IS DIMENSIONALLY COORDINATED WITH THE LADDER ELEVATIONS.

THE LADDER BLOCK IS INSERTED ON THE "S-DECK" LAYER.

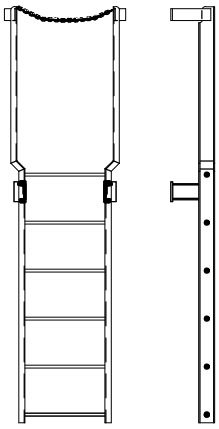




BAR RAILS



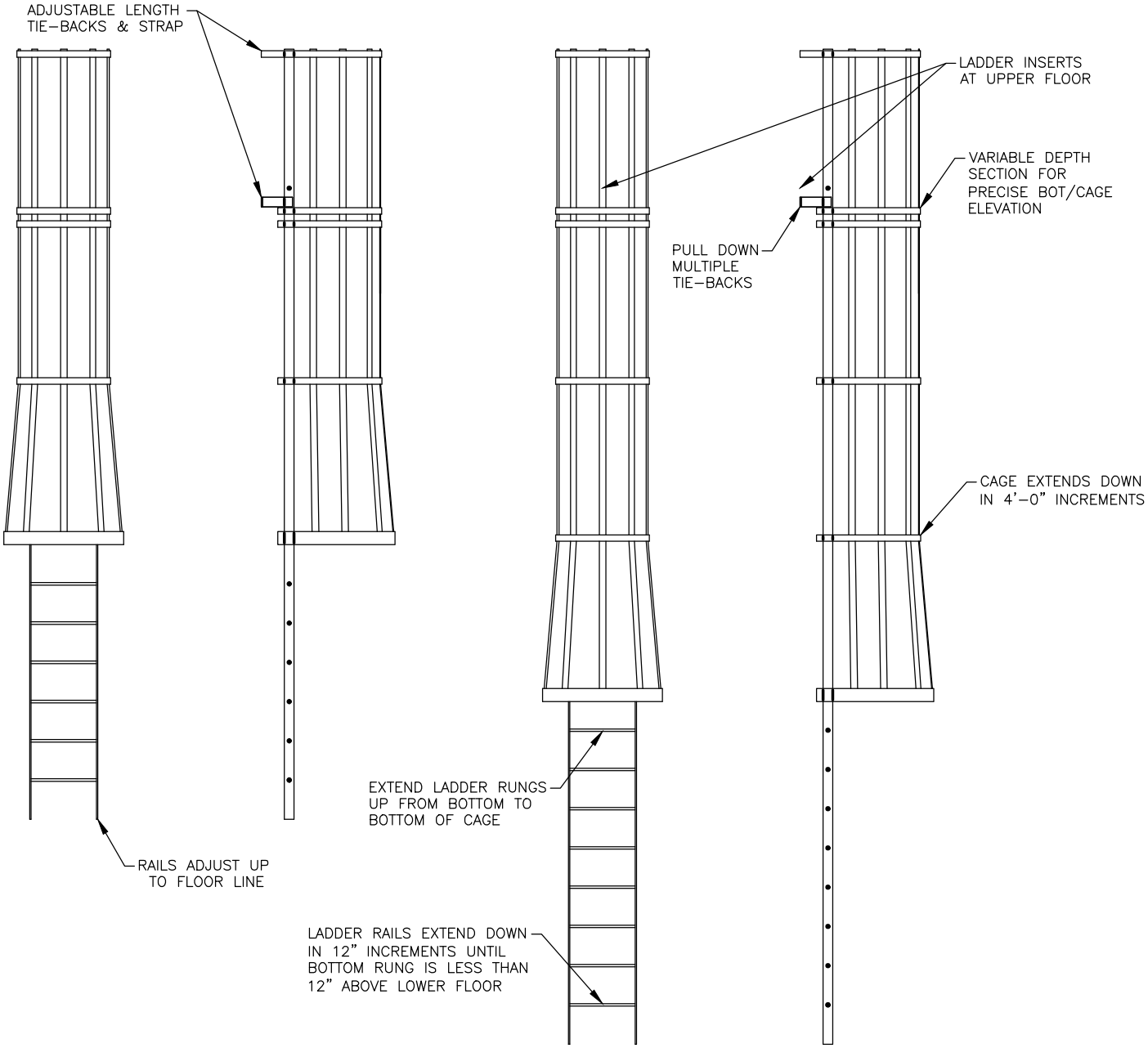
ANGLE RAILS



CHANNEL RAILS

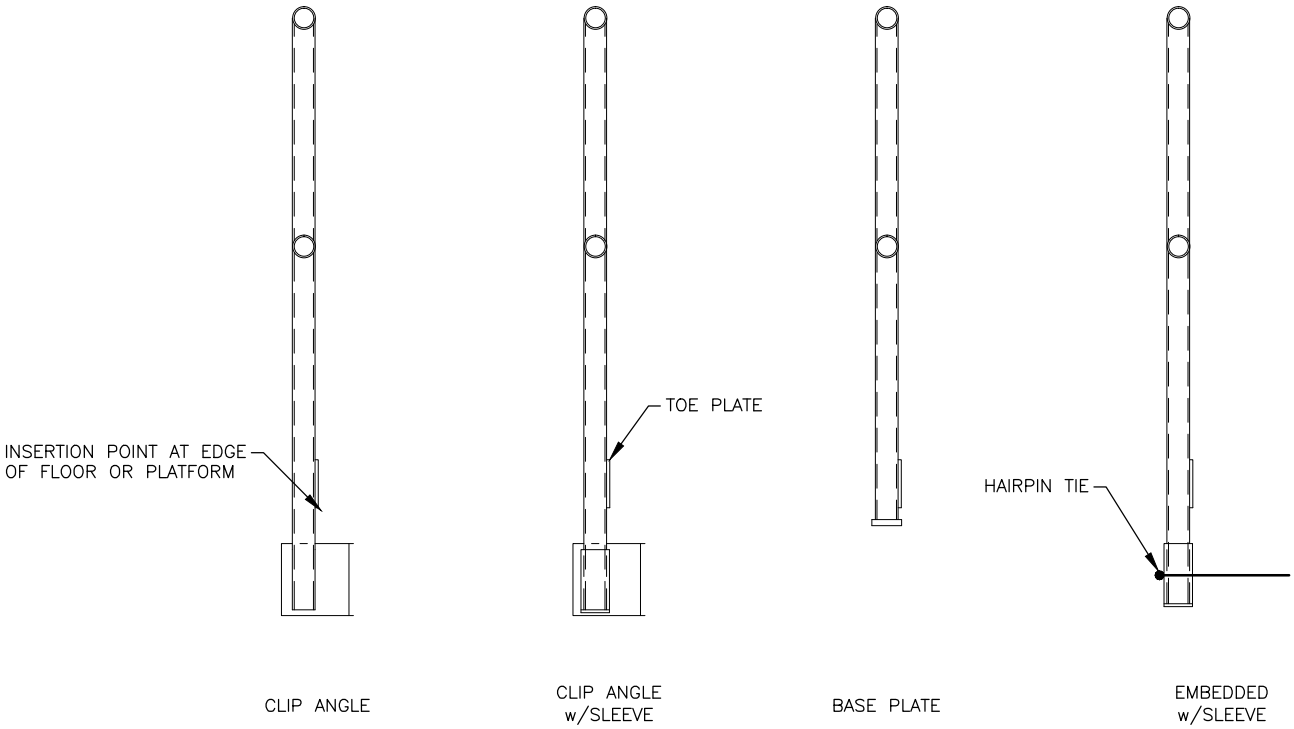
THE STEP-THRU LADDER BLOCK COORDINATES FRONT AND SIDE VIEW DYNAMICS.

THE LADDER BLOCK IS INSERTED ON THE "S-DECK" LAYER.

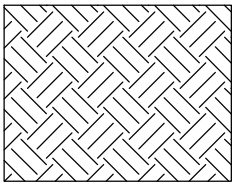


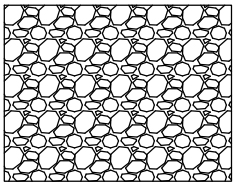
THE CAGED LADDER BLOCK COORDINATES FRONT AND SIDE VIEW DYNAMICS. IT ALSO HAS ANGLE AND CHANNEL RAIL CONFIGURATIONS NOT PICTURED HERE.

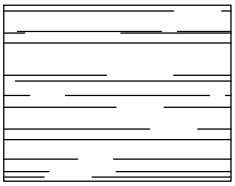
THE LADDER BLOCK IS INSERTED ON THE "S-DECK" LAYER.

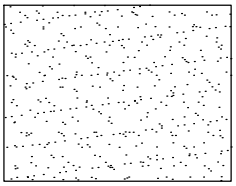


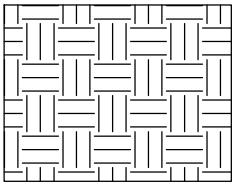
THE HANDRAIL BLOCK HAS 1½" PIPES WITH ¼"x4" TOE PLATE AND COMES WITH MULTIPLE BASES. THE HANDRAIL BLOCK IS INSERTED ON THE "S-DECK" LAYER.

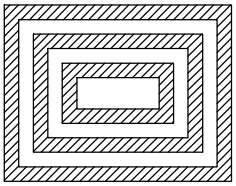

ENGINEERED FILL

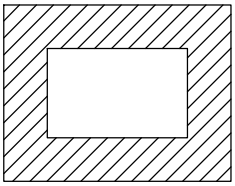

GRAVEL/POROUS FILL

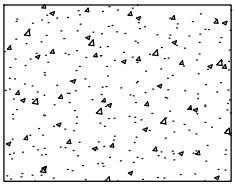

ROCK/STRATA

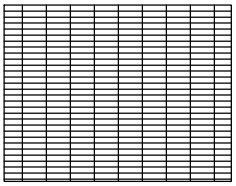

SAND

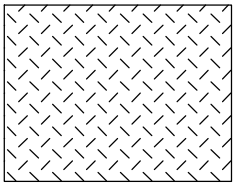

UNDISTURBED GRADE

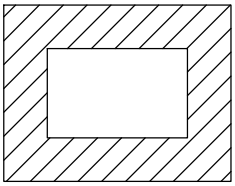

BRICK PLAN

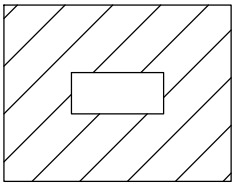

CMU PLAN

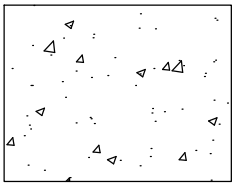

CIP CONCRETE PLAN

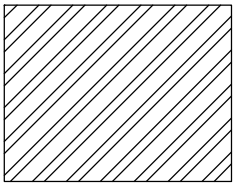

STEEL GRATING

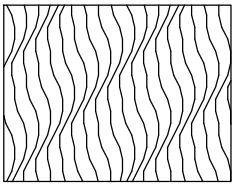

CHECKERED PLATE

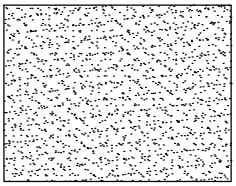

BRICK SECTION

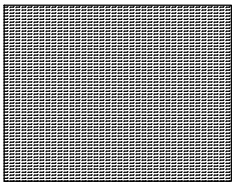

CMU SECTION


CIP CONCRETE SECTION


STEEL SECTION

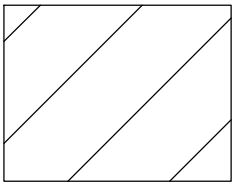

WOOD


GROUT


FLEXIBLE MATERIAL

AREA SHADING (LIGHT)

AREA SHADING (MEDIUM)


EXISTING AREA PLAN

THE HATCHING ROUTINE PROVIDED TAKES A PREDETERMINED ASSIGNMENT OF PATTERNS AND MAKES THEM AVAILABLE BASED UPON THE SCALE OF THE DRAWING BEING HATCHED. WHILE IT IS IN NO WAY A POWERFUL AS AUTOCAD'S HATCHING COMMAND, IT RETAINS THE PREVIOUS SETTINGS AND THEREFORE MAKES AN EXCELLENT "FIRST HATCH" TOOL. IT ALSO HAS A MEASURABLE EFFECT ON DRAWING CONSISTENCY.

TECHNICAL NOTES BY SECTION:

THE INTUICAD SCM BLOCKS PROVIDED REPRESENT A COMPOSITE OF SORTS. THE ACTUAL PHYSICAL CHARACTERISTICS OF THE ITEM CAN BE MANIPULATED TO THE COMPANY'S PREFERENCE AND AS LONG AS THE DYNAMICS ARE NOT DISRUPTED THE ITEM WILL PERFORM THE SAME.

SOME INTUICAD SCM BLOCKS WERE CREATED NOT TO BE EXPLODED. THIS CAN BE ALTERED BY OPENING THE BLOCK IN THE BLOCK EDITOR AND CHANGING THE SETTING, EITHER IN THAT PARTICULAR DRAWING OR IN THE LIBRARY DIRECTORY.

INTUICAD SCM BLOCKS WITH ATTRIBUTES OR TEXT ARE SCALED TO THE CURRENT DIMSCALE UNLESS NOTED.

1. DRAWING SETUP
   - A. ALL INCLUDED BORDERS CREATED ON LAYER 0 WITH COLORS ASSIGNED
   - B. PROJECT NUMBER 000000 WILL INSERT A DEFAULT OFFICE BORDER
   - C. VALID PROJECT NUMBERS WILL EXTERNALLY REFERENCE THE PREASSIGNED PROJECT BORDER FROM THE PROJECT DIRECTORY AND INSERT A BLOCK OF ATTRIBUTES.
   - D. BORDERS WITH TIGHT SPACING OR SPECIAL JUSTIFICATIONS MAY HAVE DYNAMICS ADDED.

2. SYMBOLS
   - E. IT IS HIGHLY RECOMMENDED THAT OFFICE SYMBOLS BE INCLUDED IN ANY CUSTOMIZATION PACKAGE. CROSS REFERENCING AND SYMBOLOGY ACCOUNT FOR NEARLY HALF OF THE MOST COMMON EDITS ON CONTRACT DRAWING MARK-UPS.

3. STEEL
   - F. DO NOT PUT DYNAMIC FASTENERS INSIDE OF CONNECTOR BLOCKS. DOING SO MAY CAUSE THE DYNAMICS OF THE ENTIRE BLOCK TO FAIL AS WELL AS THE DYNAMICS IN OTHER OCCURRENCES OF THE FASTENER BLOCK.
   - G. CONNECTION ANGLES DETAILED ARE GENERALLY WELDED TO THE SUPPORTED BEAM AND BOLTED TO THE SUPPORTING MEMBER.
   - H. SHEAR TABS ARE SHOP WELDED TO COLUMN ON SIDE THAT FACILITATES BEAM ERECTION.
   - I. THRU PL'S ARE SLOTTED THRU COLUMN ON SIDE THAT FACILITATES BEAM ERECTION AND WELDED ALL AROUND.
   - J. BOLT SPACING AND SIZES ARE NOT GIVEN FOR STANDARD CONNECTIONS.
   - K. BEARINGS, CLEARANCES AND WEB ANGLE FOR JOISTS ARE IN GENERAL CONFORMANCE TO SJI SPECIFICATIONS.
   - L. INTUICAD SCM JOIST BLOCKS HAVE EXTREMELY SENSITIVE SETS OF DYNAMICS. EDITING SHOULD BE DONE WITH A FULL UNDERSTANDING OF THE GEOMETRIES AND PARAMETERS.
   - M. THE JOIST AND GIRDER BLOCKS REPRESENT THE FULL SPECTRUM IN THE TABLES OF SJI.
   - N. THE WELD SYMBOLS PROVIDED A COMBINATION OF CHALLENGES. THE AWS IS ADAMANT ABOUT WELD SYMBOLS AND PLACEMENT OF INFORMATION. MOST OFFICES ARE EAGER TO GIVE UP WELDING OPTIONS THEY DO NOT FEEL THEY NEED. HAVING THE FULL SET IS ACCOMPLISHED EASILY WITH A VISIBILITY PARAMETER. THE MOMENT AUTODESK ALLOWS VISIBILITY MODES TO BE SET FROM THE COMMAND LINE THE WELDING ROUTINE WILL BE ABLE TO BE REFERRED TO AS TRULY DYNAMIC. THE OPTION OF ATTRIBUTES OR NOT FOR THE RESPECTIVE SYMBOLS IS BASED ON THE AWS.
   - O. WHERE LARGER WELDING SYMBOLS MAY CONFLICT WITH THE ATTRIBUTES, DYNAMICS HAVE BEEN PROVIDED TO ADJUST THE ATTRIBUTE LOCATION.

4. CONCRETE
   - P. THE CONCRETE REINFORCING BLOCKS AND ACCESSORIES PROVIDED REPRESENT LESS OF A COMPOSITE OF INDUSTRY STANDARDS THAN A COMPILATION OF "NOT QUITE TO SCALE" DRAFTING TECHNIQUES. ACTUAL MEMBER SIZES MAY VARY GREATLY WITH FABRICATOR, MANUFACTURER AND DESIGN CONDITIONS.
   - Q. THE CONCRETE CAP CONFIGURATIONS PROVIDED ARE FROM THE TABLES OF THE CRSI HANDBOOK OF CONCRETE DETAILING. ACTUAL CONCRETE CAP SIZES MAY VARY GREATLY WITH DESIGN CONDITIONS.
   - R. THE HELICAL PIER BLOCKS PROVIDED REPRESENT A COMPOSITE OF THE MANY TYPES AVAILABLE.
   - S. THE SHEET PILING PROFILES CORRESPOND TO ASTM.
   - T. THE BOLT GAGE DETAIL PROVIDED REPRESENTS NO SPECIFIC DESIGN WHATSOEVER. IT IS ENTIRELY AN EXERCISE IN MANAGING TYPICAL DETAILS AND DOES NOT ASSUME ANY DESIGN CAPACITIES.
   - U. THE BASE PLATE DETAIL PROVIDED REPRESENTS NO SPECIFIC DESIGN WHATSOEVER. IT IS ENTIRELY AN EXERCISE IN MANAGING TYPICAL DETAILS AND DOES NOT ASSUME ANY DESIGN CAPACITIES.

5. MASONRY
   - V. HATCHING IN BRICK AND CMU BLOCKS IS HIGHLY SENSITIVE.

6. WOOD
   - W. THE WOOD TRUSS BLOCKS PROVIDED REPRESENT NO PARTICULAR TRUSS DESIGN OR CAPACITY.

7. STAIRS/LADDERS/HANDRAILS
   - X. INTUICAD SCM STAIRS LADDERS AND HANDRAILS, AS PRESENTED, ARE IN GENERAL CONFORMANCE TO OSHA SPECIFICATIONS.